

LINGUISTICS

Language has been the most effective tool for our communication since time immemorial. Though some basic communications can be handled with gestures, language becomes inevitable for communicating elaborate and complex ideas. Scientific discovery, the advancement of knowledge, and collaborative working require a common method of communication.

Language plays this role in a civilized society. As technology and computing prowess improves, we are able to develop better applications using Artificial Intelligence (AI) techniques. This requires us to develop efficient Natural Language Processing (NLP) capabilities. A systematic study of languages and their capabilities for the evolving requirements has become important for the science and technology community.

Language is a tool used by everyone in a community and it is very difficult to maintain it unchanged. Studying the structure of the language helps us not lose the underlying principles that govern a language and ensures that the received wisdom from the ancestors is not lost. It also maintains continuity in language processing. Communication is key to trade, science and technology, and societal progress. It hinges on our ability to effectively process language as it is central to all human transactions and pursuits.

LINGUISTICS

Linguistics (L.) (from Lat. lingua – “language”) is scientific study of language.

Is a branch of language research that provides a scientific study of a language. It is a systematic study of language to understand speech sounds, grammatical structures, and meaning.

It helps us analyse the language form and meaning and identify systematic methods integral to the language to derive the word forms and their meaning using structured rules and syntax.

Linguistics (meaning)

Linguistics is the scientific study of **language and its structure**. It focuses on how humans:

- Speak and produce sounds (phonetics, phonology)
- Form words (morphology)
- Build sentences (syntax)
- Understand meaning (semantics)
- Use language in communication (pragmatics)

Linguistics is the scientific study of language. At the heart of linguistics is an understanding of:

1. What is language?
2. How does language work?
3. What do all languages have in common?’,
4. What range of variation is found among languages?
5. How does human language differ from animal communication?’,
6. How does a child learn to speak?
7. How does one write down and analyse an unwritten language?’,
8. Why do languages change?
9. To what extent are social class differences reflected in language?

AIM

Linguistic theory is the study of the nature of human language and the underlying principles that govern it. The study of language structure is a key part of linguistic theory, and linguists analyze language in a variety of ways:

1. Syntax: The study of sentence structure and how it affects meaning and interpretation
2. Semantics: The study of meaning
3. Morphology: The study of word structure and how it relates to sound and sentence structure
4. Phonetics: The study of sound systems, sound patterns, and sound structures
5. Phonology: The study of the abstract sound system of a language
6. Pragmatics: The study of how context contributes to meaning

Linguistic theory also encompasses a variety of frameworks and perspectives, including:

1. Evolutionary linguistics: The study of the origins and development of language from an evolutionary and cognitive perspective
2. Cognitive linguistics: The study of the relationship between language and cognition

Aims of linguistic theory:

The main aim of linguistic theory is twofold

1. first, to characterize what human languages are like (descriptive adequacy) and
2. second, to explain why they are that way (explanatory adequacy)

other

1. What is knowledge of language? (Competence)
2. How is knowledge of language acquired? (Acquisition)
3. How is knowledge of language put to use? (Performance/language processing)

SCOPE

The scope of linguistics is the scientific study of language, which includes many branches and subfields. The scope of linguistics is vast and includes both theoretical and practical applications:

1. Linguistic analysis; -The study of the structure and meaning of language, including syntax, semantics, morphology, phonetics, and phonology
2. Linguistic perspectives; -The study of language from a variety of perspectives, including synchronically, diachronically, and in monolinguals or multilinguals
3. Linguistic branches; -The study of language in relation to other areas, including sociolinguistics, psycholinguistics, neurolinguistics, historical linguistics, and applied linguistics
4. Linguistic applications; -The use of linguistic knowledge in real-world situations, such as language teaching

Linguistics emerged from the field of philology, and some of its branches are more qualitative and holistic in approach.

COMPONENT OF LANGUAGES

Language processing has two dimensions:

1. Receptive and
2. Productive.

RECEPTIVE SKILLS	PRODUCTIVE SKILLS
• Listening • Reading	• Speaking • Writing

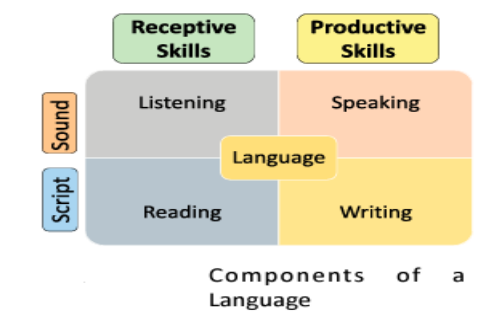
The receptive ; -The ability to understand and comprehend language through listening and reading. Is a part of a language deals with the ability of an individual to receive language inputs from multiple sources and process them to decipher the intended message and comprehend them. Receptive language involves gaining meaning from sounds, words, written information, and visual information from the environment.

Productive. ; -The ability to produce words, phrases, sentences, and paragraphs through speaking and writing. Productive skills are also known as active skills.

On the other hand, the productive part of the language is to transmit back to others for their consumption. The focus in the former is on listening and reading, Language whereas it is on speaking and writing in the latter.

Viewed from another perspective, sound (listening Reading Writing and speaking) and script (reading and writing) are the essential elements of a language. Therefore, language1 Components of a processing can be represented in a 2 x 2 framework

Linguistics address all of this aspect and phonetics will cater Receptive and Productive aspect of sound and a syntactical structure will cater to the spiritual aspect of language



BRANCHES

The core branches of Linguistics are associated with language structure and the units of each level of this structure at which Linguistics is looking at:

1. sounds,
2. words,
3. sentences, and
4. Texts.

1. If Linguistics looks at sound, then we can distinguish between the following linguistic branches:

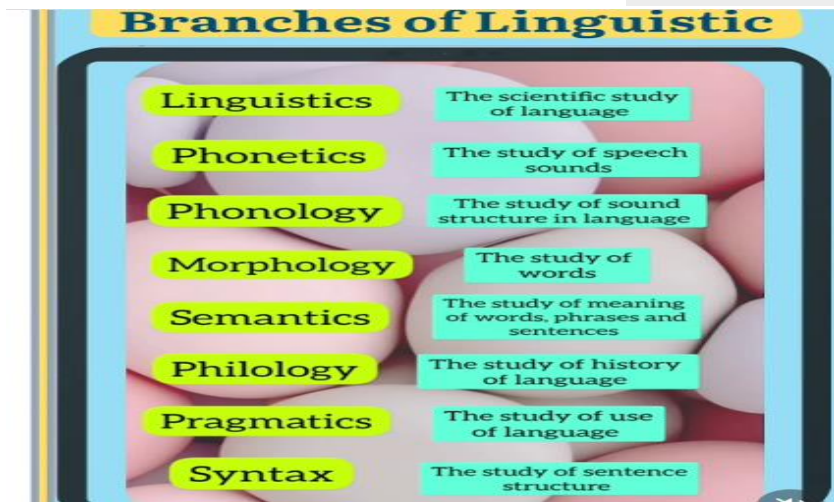
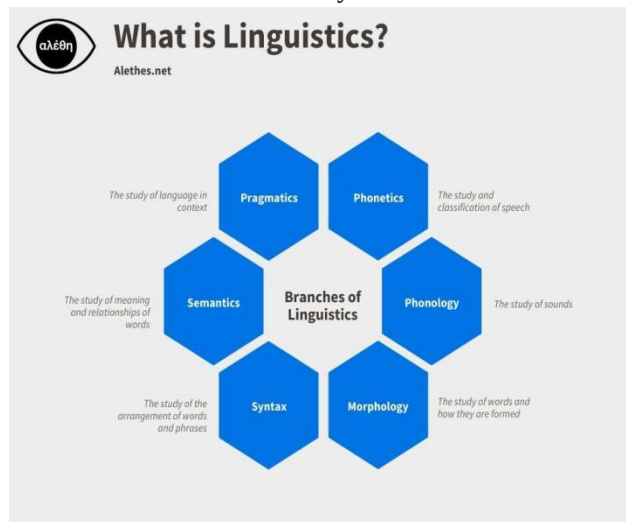
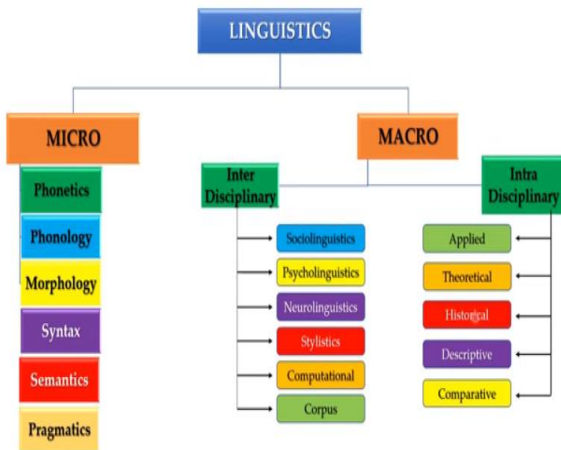
- a) **phonetics** – deals with the production (articulatory phonetics), physical transmission (acoustic phonetics), and reception or perception (auditory phonetics) of speech sounds by people (e.g. what organs of speech participate in articulating the sound and
- b) **phonology** – how sounds are organized in a particular language (e.g. how many consonants and vowels does this language have?)

2. If Linguistics looks at structure of words and sentences, we can distinguish between:

- a) **morphology** – deals with the structure of words (e.g. the word “superuser” consists of the stem ‘superuser’, two roots: ‘super’ and ‘use’, the suffix ‘-r’, and a zero ending) and
- b) **syntax** – how phrases and sentences are constructed

3. If Linguistics looks at meaning, we can distinguish between:

- a) **semantics** – has to do with meaning of words and other communicative units
- b) **pragmatics** – how meaning works to produce a particular communicative effect
- c) **discourse** – language in use – the focus here is the text and how it functions in a variety of contexts.



Micro linguistics focuses on the study of language itself, including its sound (phonetics and phonology) grammatical structures (morphology), syntax, and meanings (semantics) in context (pragmatics).

Macro Linguistics takes a broad view of linguistic phenomena, studying language in different context and its development over time. Macro-linguistics includes study of other disciplines that are connected with language study in any perspective e.g. the study of relation between society and linguistics is sociolinguistics.

Macro-linguistics is further divided into Inter disciplinary branches of linguistics and Infra disciplinary branches of linguistics.

Inter Disciplinary Branches of Macro Linguistics;- deals with study of linguistics with relation to other disciplines as sociology, psychology, neurology, geography, etc. Below are inter-disciplinary branches of macro linguistics.

Intra disciplinary branches of linguistics deal with the study of linguistics within its own discipline.

MUSIC

Sound is a vibration that travels as a wave through a medium like a gas, liquid, or solid.

It is created when an object vibrates, causing nearby particles to move and collide, which in turn makes the particles next to them vibrate.

When these vibrations reach our ears, they cause the eardrum to vibrate, and the brain interprets these vibrations as sound.

Music- pleasant sounds are called music,

Music;- an arrangement of sounds in patterns to be sung or played on instruments

HISTORY OF INDIAN MUSIC

Indian music has a long history, with roots in the Vedic Age and beyond. The history of Indian music can be divided into three main periods:

1. Ancient,
2. Medieval, and
3. Modern.

Prehistoric Period 3000 - 1800 BC

Indus Valley Culture :Mohenjo-daro, Harrappa.

Musical instruments including flutes, drums and string instruments have been found in excavations of Indus Valley sites.

Vedic Period 1500 BC

Beginning of Aryan immigration from Central Asia.

Vedas

(Sanskrit liturgical hymns, prayers, spells and incantations) :

Rig Veda - 3 note chant.

Sama Veda - 7 note chant.

Atharva Veda

Yajur Veda - poetic meters are antecedents of tala (rhythmic cycle).

Upanishads, 800 BC Brahmanas, Puranas (other Sanskrit liturgical texts).

Gandharva Gaan;- (Music of the Celestial Singers) in practice.

Buddhist music 273 - 232 BC is music (Sanskrit: vādita, saṅgīta) created for or inspired by Buddhism and includes numerous ritual and non-ritual musical forms

Reign of Emperor Ashoka 200 BC;- Panini codifies Sanskrit grammar.

Natya Shastra - by Bharata Muni 200 - 300 AD;- (treatise on the art of vocal and instrumental music, dance and drama - known as the fifth Veda).

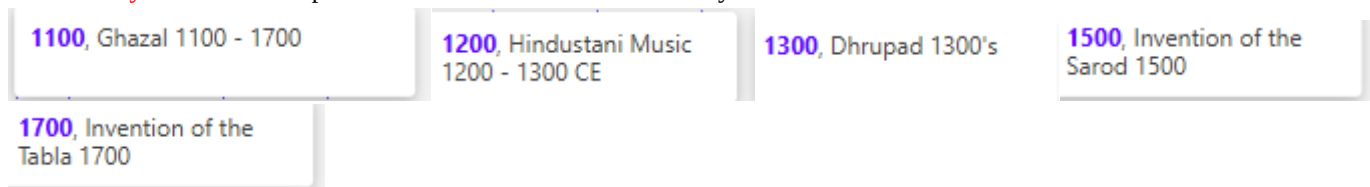
Matanga's Brihaddeshi (5th century AD–13th century AD): The term raga was first used in this work.

Gita Govinda (5th century AD–13th century AD): Jayadeva is considered a trailblazer for composing a work with structured compositions.

Sangita Ratnakara (5th century AD–13th century AD): Shargnadeva's work covered concepts like raga, talas, musical instruments, and voice culture.

12th–13th century AD: Persian and Central Asian influences began to affect the music of North India.

14th century: Indian music split into the Hindustani and Carnatic systems.



ANCIENT PERIOD

The ancient literature of our country, like the Vedas, Agamas, Upanishad, Vayu Purana, Brihaddharma Purana, Ramayana, Mahabharata, Bhagavata, Shikshagranthas and others contain invaluable references to the basic principles of classical music such as - seven swaras, three gramas, twenty onemurchanas, threelayas (speeds), nine rasas, three sthayis (octaves), srutis, etc.

Reference available dating back to the Vedic Age (2000-1000 BC) mentions a number of musical instruments. Sama Veda gives three types of notes – Udatta, anudatta and swarita, In Rig Veda – Veena, Vanshi (Flute) and Damaru

In epics the Ramayana specifies the time for the practice of the music and in Mahabharata refers to musical instruments played at the commencement of the Bharata war like sankha, bheri, anakagomukha etc.

AD 400-500 – Bharata's Natyasastra is the first and most dependable evidence to be found regarding the basis of Indian Classical music

AD 400-1100 considerable progress in music can be seen and that the jatis were subsequently consolidated into six basic melodies and then came into being two branches --- the Hindustani and Carnatic Music schools

MEDIEVAL PERIOD

India had one system of music through out the country till about 13th century. Thesame fundamentals like saptaswaras, octave, sruti etc. formed the basic principles. Haripala, for the first time mentioned the terms Hindustani and Karnatak (Carnatic) music. With the advent of muslim rule in North, the art of Indian music interacted with the Arabian and Persian systems of music. Patronised by the Muslim rulers in their royal court, the Indian music branched out to develop along with new dimension. Comparatively South India remained undisturbed without any foreign invasions or upheavals. Indian classical music continued to prosper and grow along the ancient traditional way encouraged by temples and traditional Hindu Kings. Thus Hindustani and Kamatak music developed into two independent systems of music emerging from the same, single source- Vedas.

Emperor Akbar (AD 1556-1605)... Music reached its zenith... a great connoisseur of art and culture particularly the Classical music... Thirty six experts in the art of music.. Chief among them were Tansen, Baiju Bawra and Ramdas

18TH CENTURY - THE GOLDEN AGE

During this period there was multifaceted development and musical activity, both in quality and quantity of the musical forms, Ragas, Talas, Musical instruments, musical notation system etc. The scholarly musical forms such as well decorated Kritis, Swarajatis, Varna, Pada, Tillana, Jawali, Ragamalikas etc. were composed in large numbers.

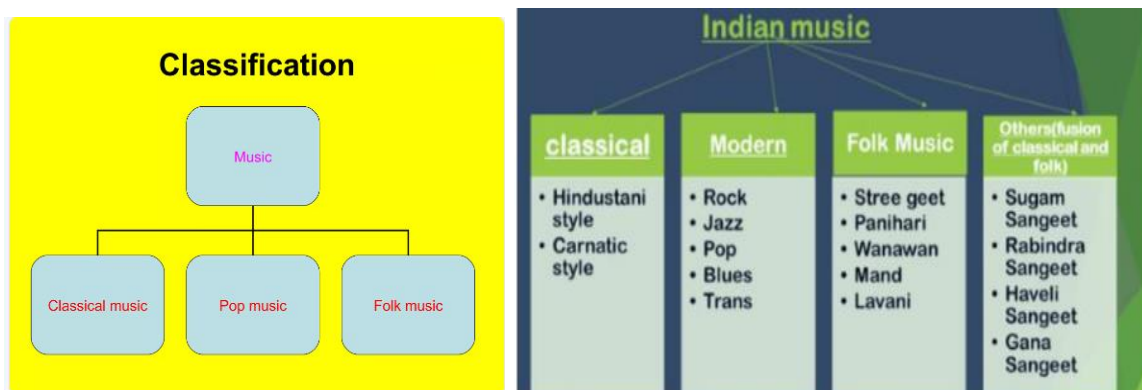
It is important to mention here that all these different forms of compositions drew their fundamentals from the ancient prabandhas. Only these sections; the musical and lyrical aspects had assumed a refined and transformed shape in the newer compositions

Classical music too started being exported out of the country in the 60's, and an experiment of combining western music with the Indian Classical form also took place.

This gave rise to what is popularly referred to as fusion music. In the 70's and 80's disco and pop music entered the Indian musical scene. The 90's further popularised the pop trend among the Indian audiences. With the further spread of information technology and an increasingly global world, we see a host of musical forms existing in contemporary India—rock, Hip-hop, jazz etc

Apart from these western forms of music, traditional forms of Indian music, such as Khayal, Ghazal, Geet, Thumri, Qawwali etc. also find place in the contemporary music. Bhajans and Kirtans, which form a separate stream of religious songs, are also quite widely sung across the country.

MUSIC CLASSIFICATION



Classical music: Also known as art music, this music is considered to be of high value and is often characterized by complexity and formality.

It is often associated with Western European churches and royal courts, and is distinct from folk and popular music.

Pop music: Modern pop music is often influenced by classical pieces.

Folk music: Folk music is often based on diverse themes and has a brisk rhythm, making it suitable for dancing. It is often characterized by a lack of hard rules.

CLASSICAL MUSIC



Two distinct classical music emerged over time:

1. Hindustani music - It is practised in northern India.
2. Carnatic music - It is a type of Indian classical music that is popular in the southern parts of the country.

North Indian Music (Hindustani Music) method is prevalent in the states of Maharashtra, Gujarat, Punjab, Rajasthan, Madhya Pradesh, Bihar and West Bengal. Majorly found in the northern part of the Indian sub-continent evolved in northern India around the 13th and 14th centuries

Hindustani music was influenced by Mughal Persian performance customs and old Hindu musical traditions, Vedic philosophy, and natural Indian sounds..Musical instruments used in Hindustani are Tabla, Sarangi, Sitar, Santoor, Flute and violin

It is based on the Raga system. The Raga is a melodic scale comprising of basic seven notes.

Hindustani Music is vocal-centric

Music from the north can be divided into two types: 1) classical and 2) light classical (also referred to as semi-classical). The classical form requires stricter adherence to the raga formula while light classical allows more opportunities for deviations and does not require the intense concentration that classical Indian music requires.

Light classical music is defined as a style of music that follows the rules of raag and taal but adheres to them less strictly than with classical music. The alaap is usually very short or doesn't exist and the melodies are often derived from popular folk music and are rendered in medium ("madhyakaal") or fast ("teevragati") tempo

There are ten main styles of singing in Hindustani music like the Dhrupad, Khayal, Tappa, Chaturanga, Tarana, Sargam, Thumri and Ragasagar, Hori and Dhamar.

South Indian Music (Karnataka Music) Carnatic music is a part of the south- Indian culture ,Carnatic music or Karnataka Sangita (known as Karnāṭakasaṃgīta or Karnāṭakasaṅgītam in the Dravidian languages) is a system of music commonly associated with South India, including the modern Indian states of Andhra Pradesh, Karnataka, Kerala, Tamil Nadu and portions of east and south Telangana and southern Odisha.

The main emphasis in Carnatic music is on vocal music; most compositions are written to be sung, and even when played on instruments, they are meant to be performed in gāyaki (singing) style.

Although there are stylistic differences, the basic elements of śruti (the relative musical pitch), svara (the musical sound of a single note), rāga (the mode or melodic formulae), and tala (the rhythmic cycles) form the foundation of improvisation and composition in both Carnatic and Hindustani music.

Although improvisation plays an important role, Carnatic music is mainly sung through compositions, especially the kriti (or kirtanam) – a form developed between the 14th and 20th centuries by composers such as PurandaraDasa, and the Trinity of Carnatic music

Types of Carnatic Music and its meaning Ragam

1. Tanam-Pallavi – Elaborate rhythmic and melodic variation in unmeasured sense.
2. Kriti-Kirthanai – Most popular type which refers to devotional music laced with poetic beauty.
3. Varnam – Performed at the beginning of a concert; a completely composed piece.
4. Padam – Slower tempoed love songs referring to the human yearning for the adored god head.
5. Javalis – Faster tempoed love songs with direct description of human love.
6. Tillana – Meaningful phrases are interspersed with variety of meaningless syllables.

Regional Music

Cultural traditions from various regions of the country reflect the rich diversity of Regional Music of India. Each region has its own particular style.

1. Tribal music
2. Folk music

Is not taught in the same way that Indian classical music is taught. There is no formal period of apprenticeship where the student is able to devote their entire life to learning the music, the economics of rural life does not permit this sort of thing. The musical practitioners must still attend to their normal duties of hunting, agriculture or whatever their chosen profession is.

Music in the villages is learnt from childhood, the music is heard and imbibed along with numerous public activities that allow the villagers to practice and hone their skills.

The music is an indispensable component of functions such as weddings, engagements, and births. There is a plethora of songs for such occasions. There are also many songs associated with planting and harvesting. In these activities the villagers routinely sing of their hopes, fears and aspirations.

Musical instruments are often different from those found in classical music. Although instruments like the tabla may sometimes be found it is more likely that cruder drums such as daf, dholak, or nal are used. The sitar and sarod which are so common in the classical genre are absent in the folk music. One often finds instruments such as the ektar, dotar, rabab, and santur. Quite often they are not called by these names, but may be named according to their local dialect. There are also instruments which are used only in particular folk styles in particular regions. These instruments are innumerable.

The instruments of classical music are crafted by artisans whose only job is the fabrication of musical instruments. In contrast the folk instruments are commonly crafted by the musicians themselves.

It is very common to find folk instruments that have been fabricated of commonly available materials. Skin, bamboo, coconut shells, and pots are but a few commonly available materials used to make musical instruments.

Main are

1. **Rasiya Coot, Uttar Pradesh** the rich tradition of singing RasiyaGeet flourished in Braj which is the sacred land of Lord Krishna's charming leelas from time immemorial.
2. **Pankhida, Rajasthan** Sung by the peasants of Rajasthan while doing work in the fields, the peasants sing and speak while playing algozaandnranjira.
3. **Lotia, Rajasthan** 'Lotia' is sung in the chaitra month during the festival - 'Lotia'. Women bring lotas (a vessel to fill water) and kalash vessel considered to be auspicious to fill water during worship) filled with water from ponds and wells.
4. **Pandavani, Chhattisgarh** In Pandavani, tales from Mahabharata are sung as a ballad and one or two episodes are chosen for the night's performance.
5. **Shakunakhar - Mangalgeet, Kumaon** Numberless songs are sung on auspicious occasions in the foothills of Himalaya. Shakunakhar are sung during religious ceremonies of baby-shower. child-birth, Ghhati (a ritual done on the sixth day from the birth of a child) Ganesh poojaetc..
6. **Mando, Goa**Goan regional music is a treasury of the traditional music of the Indian subcontinent. Mando, the finest creation of Goan song is a slow verse and refrain composition dealing with love, tragedy and both social injustice and political resistance during Portuguese presence in Goa.
7. **Alha, Uttar Pradesh**Alha, typical ballad of Bundelkhand narrates the heroic deeds of Alha and Udal, the two warrior brothers who served Raja Parmal of Majoba
8. **Hori, Uttar Pradesh** The history of Hon, its evolution and tradition is quite ancient. It is based on the love pranks of 'Radha-Krishna'. Hon singing is basically associated with the festival of Holi only.
9. **Sohar, Uttar Pradesh** Social ceremonies have, at times, served as a potent factor for intermingling of different cultures. North India has a strong tradition of singing 'Sohar' songs when a son is born in a family.
10. **Chhakri, Kashmir**Chhakri is a group song which is the most popular form of Kashmir's folk music. It is sung to the accompaniment of the noot (earthen pot) rababs, sarangi and tumbaknari (an earthen pot with high neck).
11. **Laman, Himachal Pradesh** In Laman a group of girls sing a stanza and a group of boys give reply in the song.

ELEMENTS OF INDIAN MUSIC



The basic concepts of this music includes

1. Shruti (microtones),
2. Swaras (notes),
3. Alankar (ornamentations),
4. Raga (melodies improvised from basic grammars), and
5. Tala (rhythmic patterns used in percussion).

Its tonal system divides the octave into 22 segments called Shrutis, not all equal but each roughly equal to a quarter of a whole tone of the Western music.

Both the classical music are standing on the fundamentals of The seven notes of Indian Classical music. These seven notes are also called as Saptasvara or Sapta Sur. These seven svaras are Sa, Re, Ga, Ma, Pa, Dha and Ni respectively. These SaptaSvaras are spelt as Sa, Re, Ga, Ma, Pa, Dha and Ni, but these are shortforms of Shadja ,Rishabha , Gandhara , Madhyama , Panchama , Dhaivata and Nishada respectively.

Elements of Indian music

1. Tala,
2. Raga,
3. Shruti,
4. Alankara,
5. Drone

Tala: is the rhythmic structure that defines the timing and cycle of a piece of music.

Refers to a metrical cycle of beats.

It is composed of long and short beats that are accented and unaccented.

It is a complex system with various patterns and durations, and the tala dictates the rhythmic framework for improvisation.

Is known to be the rhythmic time cycle of Indian music. It is composed of long and short beats that are accented and unaccented.

Tala means clapping of one's hand in a rhythmic beat that measures musical time. It can be clapping hands, touching fingers on the lap or striking small cymbals. Tala rarely changes with the tala. Raga and tala are the two fundamental elements in Indian Music.

There are seven talas : Ata, Dhruva, Eka, Jhampa, Matya, Rupaka and Triputatala.

Raga:-

Raga: Ragat form the melodic foundation of Indian music, serving as a framework for improvisation and composition. They are based on specific scales and patterns of notes (swaras), including microtones, and are unique to each raga.

Indian music is based on a different tonal organization called Raga. A raga is an aesthetic melodic form with peculiar ascending and descending movement. It is based on a scale of five to seven notes. Indian Tonal System SA RE GA MA PA Dha Ni

It is a powerful element of any song whereby the singer sings according to a certain motif in the song to attract the attention of the audience. Each raga has its own mood and personality which works wonders on the audience when it is performed.

Swara: It is a Sanskrit word which connotes musical notes in successive steps of the Octave. It defines the relative position of the note rather than a particular frequency. A swara is different from Shruti in the context of Indian music. Swara is also referred to as solfege of Carnatic music which has seven basic notes sa, re, ga, ma, pa,dha,ni.

Shruti:-Is the twenty-two microtones, which are used as ornamentations for the raga. This adds texture to the melody of the music.

It is a Sanskrit word which is an important concept in Indian music. It is a pitch produced either by a singer or musical instrument which is audible to the human ear

Drone:

Drone: The drone, a sustained pitch or set of pitches, provides a continuous background sound against which the melody is played.

It is often played on instruments like the tanpura and creates a harmonic foundation, particularly relevant since Indian music doesn't rely heavily on traditional harmony as in Western music.

Is a low dull monotonous sound that is continuously played throughout the composition. In the Indian musical tradition the stringed instrument called Tambura is used to play the drone.

Alankara(Sanskrit: romanized: Alāṃkāra), also referred to as palta or alankaram, is a concept in Indian classical music and literally means "ornament, decoration"

Svara (or svar)

- notes of a scale
- notes represented by syllables from Indian solfege (sa ri ga ma pa da ni sa)
- all syllables, except sa and pa, can be raised or lowered to make a full 12-tone scale
- syllables not fixed to specific pitches

Tala

- metrical or rhythmic cycle
- distinguished by: - number of beats
- the way beats are grouped into larger sections
- same tala used throughout the rhythmic section(s) of the music composition

Raga

- a melodic framework
- uses a particular scale and specific melodic movements
- tonal framework for composition and improvisation
- each raga is either associated with serious or light moods and are also associated with the time of day and season
- names of every raga does not necessarily convey a meaning

Drone

- the continuous repetition of the notes sa and pa which reinforces the tonal centre throughout the piece
- these notes are played continuously on tambura (long-neck plucked string instrument)
- strings left to vibrate so tones overlap
- modern performances: drone may be provided electronically by a drone box

SIGNIFICANCE AND IMPORTANCE OF MUSIC

1. Preserving Our Rich Heritage
2. Expressing the Soul of India
3. Fostering Bonds of Unity
4. Celebrating Our Cultural Tapestry
5. The Power Of Music In Society ; -Music is one of the most powerful forms of communication on the planet. It has the ability to unite people across cultural, language, and geographical boundaries
6. Music is known to improve cognitive function in a variety of ways. It has been shown to help with memory recall, problem solving skills, multitasking abilities, focus and concentration, spatial awareness, and more.
7. In addition to these cognitive benefits, music also has a positive impact on creativity! Learning
8. Enhance Motor Skills Coordination ; -One of the most important benefits of music is that it enhances motor skills coordination. When you learn how to play an instrument or sing properly, your muscles will become stronger and more flexible in different areas of your body
9. Increase Self Esteem and Confidence ; -Learning music can be a fun process that provides you with opportunities to express yourself creatively in ways that other activities may not allow. By singing or playing an instrument in front of others — even if they don't understand what you're saying — you'll begin to build your self esteem and confidence one step at a time.
10. Provide An Escape From Everyday Life And Stress.
11. In addition, music can be a powerful tool for personal development and emotional healing
12. Increase Self Esteem and Confidence

MUSICAL INSTRUMENTS

A musical instrument is a device constructed or modified for the purpose of making music. In principle, anything that produces sound can serve as a musical instrument.

A musical instrument is a device created or adapted to make musical sounds

Classification

Indian Classification

Musical instruments are broadly classified in 4 groups:

1. Tantu Vadya i.e. the stringed instruments
 - a. Tata i.e. plucked strings
 - b. Vitata i.e. bowed strings
2. Sushir i.e. the wind instruments.
3. Avanaddha i.e. percussion instruments, mostly the drums, made of leather.
4. Ghana i.e. percussion instruments made of metal.



Tata Vadya	Sushir Vadya	Avnaddha Vadya	Ghana Vadya
1. Yantra	1. Venu	1. Panav	1. Ghanta
2. Vallaki	2. Gomukh	2. Mridanga	2. Mani
3. Veena	3. Govishan	3. Bheri	3. Kiddiki
4. Tumba Veena	4. Shankha	4. Pesi	4. Kanchi
5. Kachapi Veena		5. Turya	5. Parivhatak
6. Mahati Veena		6. Adambra	6. Bhekhla
		7. Dignidam	7. Kalaap
		8. Mardal	8. Nupur
		9. Muraj	
		10. Pushkara	
		11. Alingya	
		12. Jharjhar	

four main categories on the basis of how sound is produced.

Four main categories on the basis of how sound is produced.

1. **The Tata Vadya or Chordophones-** Stringed instruments
2. **The Sushira Vadya or Aerophones-** Wind instruments
3. **The Avanaddha Vadya or Membranophones-** sound is produced by striking a stretch membrane
4. **The Ghana Vadya or Idiophones-** sound is produced by striking metal or clay . Played using a hammer or striker. Solid instruments which do not require tuning.

TATA VADYA - STRINGED INSTRUMENTS

TATA VADYA - STRINGED INSTRUMENTS / TATA VADYA OR CHORDOPHONES:

The stringed instruments that were played with bow, plectrum or plucked with fingers.

The tata vadya is a category of instruments in which sound is produced by the vibration of a string or chord.

These vibrations are caused by plucking or by bowing on the string which has been pulled taut.

The pitch and duration of the sound are determined by the length of the string and the amount of tension it has. Increasing the tension raises the pitch, and decreasing the tension lowers it.

The length of the vibrating string directly affects the pitch of the note. Shorter strings produce higher pitches, and longer strings produce lower pitches.

The tata vadya are divided into two broad categories

the plucked and

the bowed,

and further subdivided into the fretted and non-fretted variety.

I. Plucked String Instruments (Tat) - These instruments are played by plucking or striking the strings.

Sitar, Tanpura, Veena, Vichitra Veena, Saraswati Veena, Ektara, Sarod, Surbahaar, Surmandal, Rebab, Santoor etc. are examples of
of Plucked String Instruments.

ii. Bowed String Instruments (Vitat) - These string instruments are bowed.

Bowed string instruments are a type of musical instrument where the strings are made to vibrate by a bow, causing sound to be produced. Examples include the

Chikara, Dilruba, Sarangi, Ravanhasta, Taar Shehnai, Israj etc. are examples of Bowed String Instruments

Veena, Kand Veena, Karkari Veena are mentioned.

fretted instruments have metal bars (frets) along neck marking specific note location like guitar, basses, mandolins

Unfretted instruments have uninterrupted finger board like violin, cellos

Mechanism of music in Tatyavada (chordophones)

Tatyavada instruments are stringed instruments where sound is produced by the vibration of a string or chord.

Plucked strings (Tat): Sound is produced by plucking or striking the strings, causing them to vibrate

.Mechanism: The vibration from the string is transferred to the bridge and then to the instrument's body, which amplifies the sound by moving a larger area of air.

Initial vibration: The musician creates a vibration in one or more taut strings by plucking, striking, or bowing them.

Controlling pitch: The pitch is determined by several factors related to the vibrating string:

Tension: How tightly the string is pulled.

Length: The length of the string that is allowed to vibrate.

Thickness: The thickness of the string itself.

Bowed strings (Vitat): Sound is created by drawing a bow across the strings.

Mechanism: The bow's friction causes the strings to vibrate continuously. The instrument's resonating body amplifies the sound.

Pitch control: Similar to plucked strings, pressing the string against the fretboard changes the length and, therefore, the pitch.

Resonance: This vibration is transferred to a soundbox or resonator, which amplifies the sound. This hollow, resonating body is what gives the instrument its unique tone and volume.

SUSHIRAVADYA

Are wind instruments

Sushir Vadya or aerophones are wind instruments.

In the Sushira Vadya group, sound is produced by blowing air into a hollow column

These instruments use air as the primary vibrating medium for producing sound

produce sound by causing a column of air to vibrate.

Air Vibration: Sushir Vadya instruments rely on the vibration of air to generate sound. The player's breath or a mechanical system forces air into a hollow tube or chamber.

Vibrating Air Column: As air moves through the instrument, it sets a column of air within the tube into vibration.

Pitch Control: The pitch of the note is determined by the length of the vibrating air column. This length can be modified by opening and closing finger holes or by other means, such as changing the pressure of the air or using valves,

Resonator: The tube itself acts as a resonator, amplifying the vibrations of the air column

The simplest of these instruments is the flute.

These categories are based on the distance between the blow hole and the first finger hole

Key characteristics

Hollow;- the instruments has hollow body creating a resonating chamber for the air

Pitch control;- the pitch is altered by changing the length of vibrating air column, which is achieved by opening and closing the hole on the instruments surface

Fingering;- fingers are used to cover and uncover the holes, modifying the pitch and producing deferent notes

example;- flutes, shenai

Mechanism of music in Susir Vadya (aerophones)

Susir Vadya are wind instruments where sound is produced by causing a column of air to vibrate within a hollow tube.

Mechanism: Sound is made by the vibration of an air column within the instrument

Initial vibration: A musician blows air into or across an opening in the instrument, setting the air column inside into vibration.

Controlling pitch: The pitch of the note is controlled by altering the length of the vibrating air column.

Pitch control: The pitch is changed by using fingers to open or close holes along the length of the instrument, altering the length of the vibrating air column

This is achieved by:

Finger holes: The musician opens and closes holes along the instrument's body with their fingers.

Mouth position: For flutes like the bansuri, a specific blowing technique is used.

Reeds: In instruments like the shehnai, the air causes a reed or reeds to vibrate.

Types of air supply:

Mouth-blown: The most common type, where the musician provides the air directly (e.g., bansuri, shehnai).

Mechanical: Wind is supplied mechanically, as in a harmonium

AVANADDHAVADYA

In the AvanaddhaVadya category of instruments, sound is produced by striking the animal skin which has been stretched across an earthen or metal pot or a wooden barrel or frame.

Avanaddh part refers to that which is covered

- Sound is generated when stretched membrane is struck, plucked or stroked, causing vibration that produce sound wave
- A covered hollow vessel generates bits when struck
- It can play with hands, sticks or even combination of both
- Drums have been divided into different categories on the basis of their shapes and structure as also the position and placement for playing.
- The main categories are-Oordhwaka, Ankya, Alingya and the waisted or the Damaru family of drums.

PERCUSSION INSTRUMENTS:

(a) Ghata, terracotta vessel was the first percussion instrument

(b) Kanjira was first instrument with skin cover

(c) Single side: Drums, Double Side: Dhol&Mridanga

(d) Tabla and many other variants

In these instruments, body of the musical instrument is made up of special clay, wood or metal;

1. Barrel shaped with both the sides open in the opposite ends e.g., Mridanga, Pakhawaja, Dholak, Madal etc.
2. Kettle shaped musical drums with only one open end e.g., Tabla, Bayan, Urdhwaka etc. is covered with best skin with the help of thin and long leather straps to keep the musical drum in perfect tone.

These are instruments that have a membrane stretched over a cylinder, the body of which can close or open at various distances from the stretched skin.

In the Avanaddha Vadya category of instruments, sound is produced by striking the animal skin which has been stretched across an earthen or metal pot or a wooden barrel or frame

Stretched membranes (Avanaddha Vadya): Sound is produced by the vibration of a stretched skin or membrane.

Mechanism: The performer strikes the membrane with their hands, fingers, or a stick, causing it to vibrate and produce sound.

Sound Production: The sound is generated by striking the membrane with hands, sticks, or other objects, causing it to vibrate. The vibration transfers to the hollow vessel, which then amplifies and resonates the sound.

Tuning: The pitch of the sound can be adjusted by varying the tension of the membrane or by using tuning mechanisms like the ones found in the tabla.

It can play with hands, sticks or even combination of both

Drums have been divided into different categories on the basis of their shapes and structure as also the position and placement for playing.

The main categories are-Oordhwaka, Ankya, Alingya and the waisted or the Damaru family of drums.

GHANA VADYA

The earliest instruments invented by man are said to be the Ghana Vadya.

These are idiophones which make a sound when struck or shaken. They are solid instruments that do not require tuning

Are solid musical instruments that produce sound through vibration when struck, shaken or scraped

Once constructed, this variety of instrument do not need special tuning prior to playing. And often used for rhythmic accompaniment

Example majira, khartal, kanchi, jhani, jaltrang,

Key characteristics;

Solid instruments;- they are made from solid material like wood, metal or other hard substance

No tuning required;- they produce sound through their inherent vibration when played, rather than needing to be tuned in specific way

Rhythmic emphasis;- they are often used to create rhythmic patterns and accompany other instruments in music

In early times these instruments were the extension of the human body such as sticks, clappers, rods, etc. and were also closely related to objects of utility in daily life such as pots and pans, jhani, falam, etc.

They are principally rhythmic in function and are best suited as accompaniment to folk and tribal music and dance.

Key characteristics;

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In these instruments, sound is produced by striking instruments made upof metal or wooden pieces. Thus, these instruments are also called metallicinstruments. It includes Jhanjh, Kartal, Manjira, Chimta and Talam, etc.They are principally rhythmic in function and are best suited as accompaniment to folk and tribal music and dance.

PERCUSSION INSTRUMENTS:

- (a) Ghata, terracotta vessel was the first percussion instrument
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- (c) Single side: Drums, Double Side: Dhol&Mridanga
- (d) Tabla and many other variants

Mechanism of music in Ghan Vadya (idiophones)

Ghan Vadya are solid, percussive instruments that produce sound through the vibration of their entire body, without the use of strings or membranes.

Solid instruments (Ghana Vadya): Sound is made by the vibration of the solid body of the instrument itself.

Mechanism: The instrument is struck or shaken, causing the material to vibrate and create sound.

Initial vibration: The sound is created by striking, shaking, or scraping the instrument. The instrument's inherent elasticity causes it to vibrate.

Fixed pitch: Most Ghan Vadya are unable to produce a wide range of definite pitches for melodic purposes, which is why their use is limited in classical music. They are primarily used for rhythmic accompaniment.

Source of sound: The material of the instrument itself—**such as metal, wood, or ceramic—is the source of the sound, and the quality depends on its composition and shape**

DHVANI SIDDHANTA

- Anand Vardhan (Kashmiri Pandit) supported Bharata's rasa theory and Natya shastra.
- He is credited with creating the **dhvani theory**.
- He wrote of dhvani (meaning sound, or resonance) in regard to the "soul of poetry.
- Dhvani is used for-
 - Conventional meaning.
 - The sound structure of words.
 - The suggestible meaning itself.
 - The power of the word to convey the suggestible meaning.
 - The poetic work containing the suggestible meaning.

DHVANI SIDDHANTA

- In Dhvanyaloka for the first time we hear the word 'dhvani' used in a technical sense of "a particular type of poetry".
- According to Anandvardhana, words can convey also a suggested meaning in addition to their conventional primary meaning.
- In the domain of poetry the suggested meaning occurs in three forms, viz, '
 1. Vastumatra' (mere matter of fact),
 2. alankar (figures of speech) and
 3. 'rasadi' The (rasa and other such mental states).
- And a composition where a suggested sense predominates is called 'dhvani'.
- In other words the suggestive aspect of poetry is called 'dhvani'.
- Thus the Dhvani Siddhanta is founded upon a threefold division of meaning, emanating from a threesome of word-powers.
 1. The first power of word is called 'abhidha' and the meaning flowing there from is called 'abhideyatha' or 'vacyartha'.
 2. The second power is called 'Laksana' and the meaning derived ther efrom, is called 'Laksyartha'.
 3. The third word-power is known as 'vyanjana' and the meaning, obtained there from, is called 'vyangyartha' or 'pratiyamanartha'.
- अभिधा (Abhidha) - Literal Meaning of the word
- लक्षणा (Lakshana) - Characteristical Meaning of the word
- व्यञ्जना (Vyanjana) - Poetic & Metaphysical Meaning of the word

Dhvani theory is a theory of meaning , of symbolism and poetry of suggestion is the highest kind of poetry. Dhvani is the method or means of achieving or evoking Rasa, which is the effect of suggestion.



Dhvani Theory:
Theory of Meaning of Symbolism - Symantic Theory

- Soul of poetry
- Importance to knowing suggested meaning of discourse
- Expression consist of words(shabda) and explicit meaning.
- Suggested sense predoninates over literal sense
- Suggestion : Soul of poetry
- Pratibha-creativity intuition, sensitivity required on the part of writer and reader

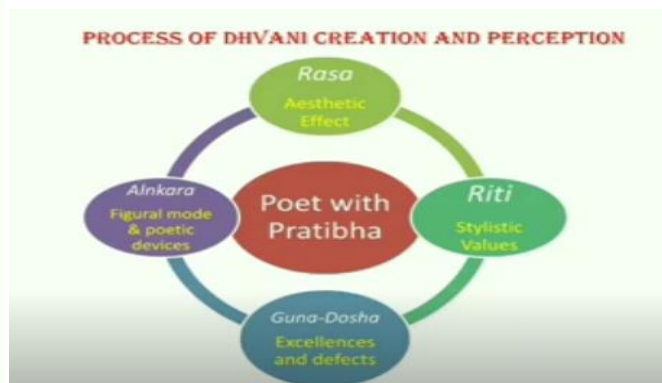
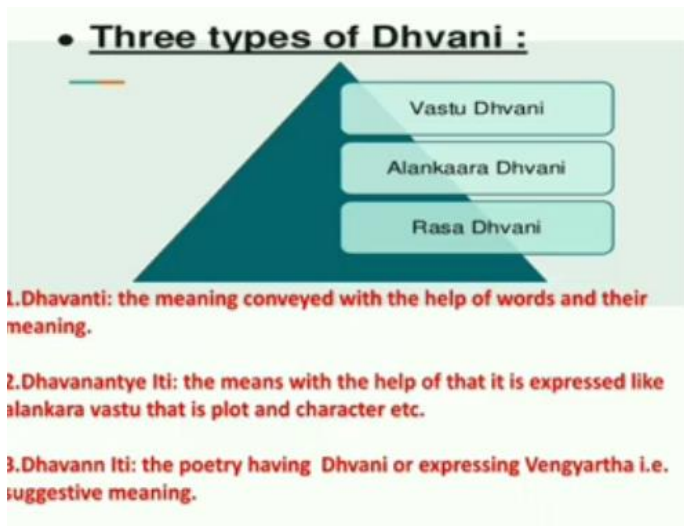
WORDS CONVEY THREE TYPES OF MEANING

- Anandvardhan 's three levels Meaning :



According to Anandvardhana, Dhvani has three aspects as

- 1) Abhidha- It consists in the literal meaning of the expression
- 2) Laksana- It consists in the external characteristic of the expression which are indicative of something deeper
- 3) Vyanjana- It indicates what is suggestive.



Dhvani Theory

A composition where suggested sense predominates is called Dhvani.

The word Dhvani is used for both - the suggestive word and the suggestive meaning.

The poetic language can never be possible without Dhvani. In Dhvanyāloka, Dhvani becomes an all-embracing principle that explains the structure and function of the major elements of literature - the aesthetic effect (Rasā), the figural mode and devices (Alamkāra), the stylistic values (Riti) and excellence and defects (Guna-dosa).

Core Idea of Dhvani Theory

A literary expression has three levels of meaning:

1. **Vachya (Literal meaning)**
 - Direct dictionary meaning of words
2. **Lakshya (Implied meaning)**
 - Meaning inferred from context
3. **Dhvani (Suggested meaning) ★**
 - Deepest aesthetic meaning
 - Hidden, emotional, or symbolic sense
 - Considered the **soul of poetry (Kavya Atma)**

DANCE

Dance is a type of art that generally involves movement of the body, usually rhythmic and to music, performed in many different cultures and used as a form of expression, social interaction and exercise or presented in a spiritual or performance setting.

Natya Shastra, the ancient Indian treatise attributed to Bharata Muni, dance (Natya) is defined as a holistic, spiritual art that combines drama, music, and rhythmic movement.

HISTORY OF DANCE

The history of dance in India is one of the oldest and richest cultural traditions in the world. Indian dance developed over thousands of years through religion, mythology, temple traditions, royal courts, and folk culture.

Origin of Dance in India

- Indian dance began in ancient times as a form of worship and storytelling. Early humans expressed emotions through body

HISTORY OF DANCE

- **Evidence from Indus Valley Civilization**
- Archaeologists found the famous bronze statue called the “Dancing Girl” from Indus Valley Civilization (around 2500 BCE), showing that dance existed in India thousands of years ago.
- **Dance in Vedic Period**
- During the Vedic age, dance was connected with rituals, festivals, and religious ceremonies. The four Vedas mention music and dance:
 - **Rigveda** – songs and hymns
 - **Samaveda** – music
 - **Yajurveda** – ritual movements

- **Atharvaveda** – cultural practices
- Dance was considered a sacred art.
- The foundation of Indian classical dance is the ancient Sanskrit text *Natya Shastra* written by Bharata Muni between 200 BCE and 200 CE.
- It explains:
 - Body movements
 - Facial expressions
 - Hand gestures (mudras)
 - Emotions (rasas)
 - Stage performance
 - Music and drama
- **Temple Dance Tradition**
 - Between the 4th and 12th centuries, dance flourished in Hindu temples. Dancers performed before deities as an act of devotion.
 - Important temple traditions:
 - Devadasi tradition in South India
 - Mahari tradition in Odisha
 - Temple sculptures also show detailed dance poses.
- **Dance During Mughal Period**
 - During the Mughal era, dance evolved in royal courts. Kathak became popular in palaces and included Persian influences, graceful spins, and expressive storytelling.
- **British Period**
 - Under British rule, many traditional dances declined because temple dance traditions were discouraged. However, Indian artists and cultural reformers later revived classical dance forms in the 20th century.
- **Post-independence**, traditional dances were promoted as part of India's heritage, and institutions were established to teach and preserve them.
- In the post independence period, there have been considerable changes in Indian classical dance styles in regard to presentation, makeup, dress, use of musical instruments, light and sound



History of Dance in India



Dance in India is as old as Indian civilization itself. It has been an integral part of our culture, tradition, rituals and social life. Indian dance is the perfect blend of art, culture, and spirituality.



1. Indus Valley Civilization (3300–1900 BCE)

- The famous 'Dancing Girl' figurine from Mohenjo-Daro is considered one of the earliest representations of dance in India.
- Dance probably formed a part of social and religious life.



2. Vedic Period (1500–500 BCE)

- Dance was a part of religious rituals and sacrifices.
- Mention of dance in Rigveda and other ancient texts.
- Women called 'Nartakis' used to dance in celebrations.





3. Ancient Period (500 BCE–1200 CE)

- Dance developed as a classical art.
- Mention in *Natyashastra* by Bharata Muni (2nd century BCE – 2nd century CE).
- Temples became centers of dance; Devadasi tradition began.
- Different dance forms like Bharatanatyam, Odissi, Kuchipudi have their roots in this period.



4. Medieval Period (1200–1700 CE)

- Dance flourished in royal courts and temples.
- Influence of Bhakti movement—dance used as a medium of devotion.
- Kathak evolved in North India.
- Devotional dances like Rasa Lila, Yakshagana developed.





5. Colonial Period (1700–1947)

- Traditional dance faced decline due to colonial influence.
- Devadasi system was criticized and gradually banned.
- Many dance forms were confined to temples and royal families.
- Efforts were made by Indian reformers to revive these art forms.



6. Modern Period (1947–Present)

- Revival and reformation of classical dances.
- Establishment of institutions like Sangeet Natak Akademi (1953).
- Dance gains national and international recognition.
- Fusion, contemporary and experimental dances emerge.
- Dance continues to evolve while preserving its rich heritage.



From ancient rituals to modern stages, Indian dance reflects the soul of our culture—expressing devotion, emotion, story and celebration.

COMPONENT OF DANCE/ ELEMENTS OF DANCE

- In the Natyashastra the dance is divided into Three basic categories;
- **Nritta** – pure dance movements
- **Nritya** – expressive dance
- **Natya** – dramatic storytelling
- **Nritta = Advas(BASIC DANCE MOVEMENTS) + Hastas (HAND GESTURES)**
- **Nritya - Advas(BASIC DANCE MOVEMENTS) + Hastas (HANDGESTURES) +Abhinaya (art of expression)**
- **Natya: = Abhinaya (art of expression)+ character potrayl (the way a character is represented)**

1. **Nritta is a term in Indian classical dance that refers to "pure dance"—abstract, rhythmic, and technical movements without any storytelling or emotion.** Nritta means pure dance, a presentation of rhythm through graceful movement of the body

Nritta (Nṛtta) is pure dance without any emotional expression. It is characterized by its rhythmic patterns and complex footwork. Nritta is often used to showcase the technical skill of the dancer

2. **Nritya or dance;-** in Indian classical dance, it specifically refers to the expressive aspect of dance that combines rhythm with emotion to convey meaning. .
 - Expressive dance or mime. It combines rhythmic movement with facial expressions, hand gestures (*Mudras*), and body language to interpret the lyrics of a song and convey specific emotions.
 - **Nritya;-** Also known as abhinaya, this aspect of dance is slower and more expressive. Nritya (Nṛtya) is an expressive dance that combines nritta with emotional expression .It uses gestures, facial expressions, and eye movements to convey a story or mood.
 - Nritya is often used to tell stories or express ideas.
 - **Nritya is also often called abhinaya**

3. **Natya means** "drama" or "the art of performance" and is a Sanskrit word for traditional Indian performing arts that

Natya;- is the highest form of dance, which combines nritta, nritya, and abhinaya. It often involves a group performance, but can also be performed solo. The dancer uses standardized body movements to indicate different characters in the story.

Natya is often used to create a theatrical experience that engages the audience on an emotional level. This can be done in two modes of presentation:

- natyadharmi (stylistic depiction) and
- lokadharmi (realistic depiction).

4. **Rasa (Emotions/Aesthetic Experience);-Rasa means the emotional effect created in the audience.**

5. **Bhava (Expression or Feeling): Bhava is the expression shown by the dancer to create rasa in the audience.**

6. **Abhinaya (Modes of Expression) Abhinaya means communication of emotions and ideas.**

7. **Tala (Rhythm) Rhythmic cycle of music and dance. Controls timing and coordination.**

8. **Laya (Tempo) Speed of performance.**

9. **Mudras (Hand Gestures) Symbolic hand gestures used to convey meanings, objects, emotions, and actions.**

The four elements of Natyashastra consist of,

1. **Aṅgika** (body movements)
2. **Vācika** (verbal expressions)
3. **Āhārya** (costumes, makeup, and props)
4. **Sāttvika** (emotional expressions).

AṅGIKA includes facial expressions, hand gestures, and body postures,

Expression through the body: This is the most fundamental type of expression in dance and involves the entire body to convey meaning.

Includes: Hand gestures (*mudras*), facial expressions, body postures, and the movement of specific parts of the body like the head, arms, and feet

VĀCIKA encompasses dialogue, songs, and narration.

Expression through speech and song: This involves the words spoken or sung during a performance.

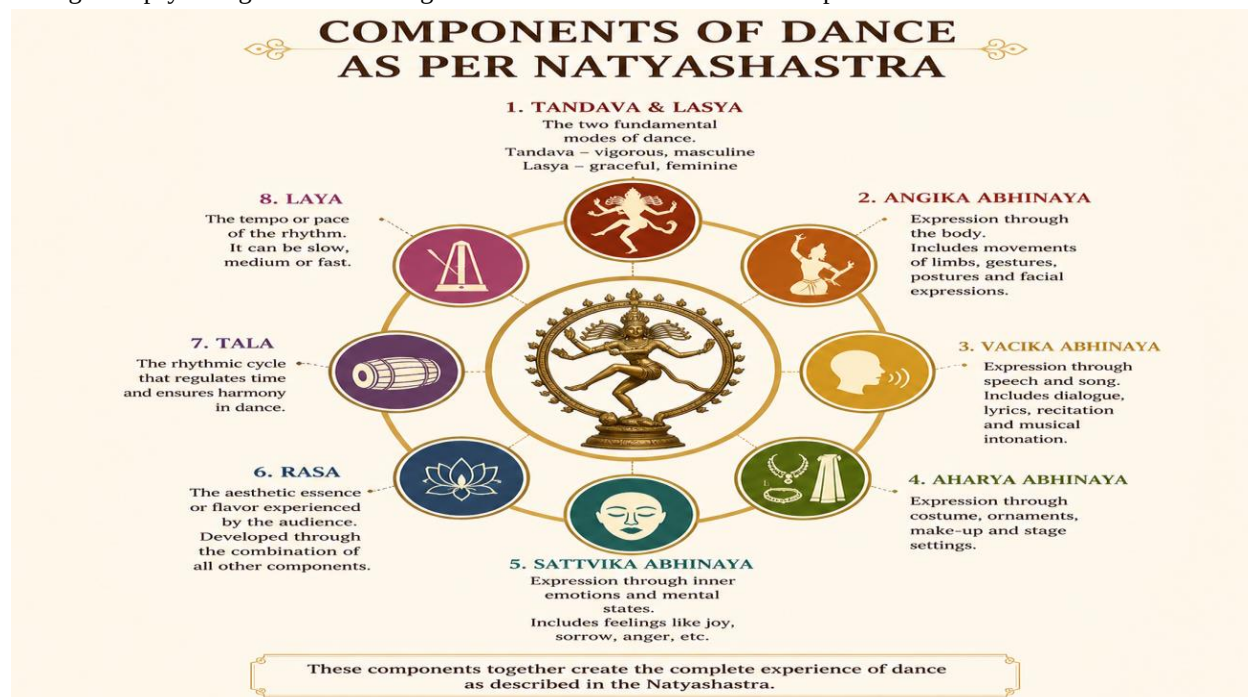
Includes: Vocalization and the use of language to move the story forward and convey emotion

ĀHĀRYA involves costumes, jewelry, makeup, and props used by actors to enhance their characters.

Expression through appearance: This refers to all the visual elements used to create a character or setting

SATVIKA ABHINAYA (Emotional Expression);-This is where the dancer transcends physicality and enters the realm of pure emotion, using trembling, sweating, or other physical manifestations to portray inner feelings.

Expression through inner emotion: The fourth component of abhinaya is Sattvika, which involves conveying the character's true feelings and psychological states through sincere and authentic emotional expression



NAVARASAS

Rasa, “taste” or “essence”, refers to the sentiment that the *bhava*, manifested by the actor, should evoke in the audience. Rasas, also known as the Navarasas, are integral to classical dance and Indian aesthetics.

Each Rasa represents a specific emotion or sentiment that can be expressed through facial expressions, body movements, and gestures.

Here are some of the key concepts of the NāṭyaŚāstra:

The Natyashastra only mentions eight rasas, and the ninth, Shanta, was added by Abhinavagupta, a philosopher from 1000 AD.

In the Natyashastra, the Navarasa are the nine emotions that are expressed in Indian classical dance:

1. Sringaram: Love and beauty
2. Hasya: Laughter and mirth
3. Karunam: Compassion and mercy
4. Bibhatsam: Disgust and aversion

5. Bhayanakam: Horror and terror
6. Veeram: Heroic mood
7. Adbhutham: Wonder and amazement
8. Raudra: Anger
9. Shanta: Peace or tranquility

By skillfully evoking these Rasas, classical dancers bring depth, emotional resonance, and a powerful connection to their performances, captivating audiences and allowing them to experience the full spectrum of human emotions.

The colors associated with the nine rasas or emotions in the Natyashastra are:

1. Shringara (Love)- Light Green: Shringara, the rasa of love, beauty, and attraction, is often represented by a soft green. This color conveys freshness, life, and renewal, embodying the harmony and beauty associated with romance and affection.
2. Hasya (Laughter) - White: White represents Hasya, the emotion of joy and laughter. As a symbol of purity and lightness, white reflects the positive, uplifting, and innocent nature of humor and happiness.
3. Karuna (Compassion) - Grey: Compassion is conveyed by grey, signifying sadness and empathy. The subdued tone of grey reflects introspection and sorrow, fitting for this rasa, which is centred on empathy and shared suffering.
4. Raudra (Anger) - Red: Red is the colour of rage, power, and intensity, ideal for Raudra. This strong and vibrant colour symbolizes aggression, strength, and fury.
5. Veera (Courage) - Orange: Courage, heroism, and valour are represented by orange. This colour reflects energy, enthusiasm, and bravery, qualities essential to Veera.
6. Bhayanaka (Fear) - Black: Black is used to symbolize fear and the unknown, as it evokes a sense of darkness, mystery, and danger. Bhayanaka taps into this sense of dread and uncertainty.
7. Bibhatsa (Disgust) - Blue: Disgust or aversion is often associated with blue, especially shades that seem cold or unappealing. This color signifies repulsion, evoking an instinctive reaction to avoid what is unpleasant.
8. Adbhuta (Wonder) - Yellow: The rasa of wonder and amazement is symbolized by yellow. Bright and radiant, yellow captures the feeling of surprise, curiosity, and discovery.
9. Shanta (Peace) - Light Blue: Peace and tranquility are best represented by light blue, which reflects calmness and serenity. Shanta embodies stillness, a release from worldly chaos.

Each colour in the Navarasas helps dancers and performers bring these emotions to life, deepening the impact of traditional Indian art forms

MUDRA

One of the most striking features of Indian classical dance is the use of hand gestures. The Natyashastra book goes on to further explain the position of the hands, palm and fingers in more detail. These positions are called MUDRAS and convey various meanings in the dance

Speaking in dance via gestures, rather than orally, in order to visually convey outer events or things, as well as inner feelings, two classifications of specific traditional 'MUDRA' (hand/finger gesture) are used in Indian Classical Dance. A dancer has to strictly comply with all the

postures, expressions and mudras in his / her dance performance.

To convey inner feelings, two classifications of mudras

1. hand or
2. finger gesture

are used in Indian classical dance,

The Abhinaya Darpa (a descriptive primer for dancers) mentions that the dancer should sing the song by the throat, express the meaning of the song through hand gestures, show the state of feelings in the song by eyes, and express the rhythm with his or her feet.

From the Natya Shastra, a text on the arts, this quotation and translation is often quoted by Indian classical dance instructors:

"Yatohastastatodrishtih..." "Where the hand is, the eyes follow"

"Yatodrishtistatomanaha..." "Where the eyes go, the mind follows"

"Yatomanastatobhavaha..." "Where the mind is, there is the feeling"

"Yatobhavastatorasaha..." "Where there is feeling, there is mood/flavour, sweetness (i.e., appreciation of art; aesthetic bliss)"

In Bharatanatyam, the classical dance of India performed by Lord Nataraja, approximately 51 root mudras (hand or finger gestures) are used to clearly communicate specific ideas, events, actions, or creatures

Asamyuta Hasta in which require only one hand, 28 mudras, There are 24 Asamyuta Hasta according to Natyashastra of Bharatmuni. They are: Pataka, Tripataka, Kartarimukha, Ardhachandra, Arala, Shukatunda, Mushti, Shikhara, Kapitha, Khatakamukha, Soochyasya, Padmakosha, Sarpasheersha, Mrigasheersha, Kangula, Alapadma, Chatura, Bhramara, Hamsaya, Hamsapaksha, Samdamsha, Mukula, Urnanabha, Tamrachuda.

'Samyuta Hasta'; - mudras which require both hands, Gestures of combined hands are thirteen in number : Anjali, Kapota, Karkta, Svastika, Kataka.- vardhamanaka, Utsanga, NishdaDola, Puspaputa, Makara, Gajadanta, Avahittha and Vardhaman

Before learning the hand gestures, here is the nomenclature of each finger as per Natyasastra.

1. Angushta – Thumb
2. Tarjani – Forefinger
3. Madhyama – Middle finger
4. Anamika – Ring finger
5. Kanishta – Little finger

General rules regarding the use of hand gestures

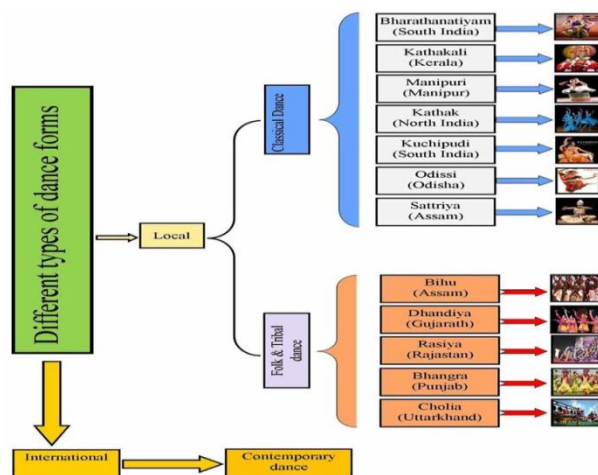
1. In acting, hand [gestures] should be selected for their form, movement, significance, and class according to the personal judgment [of the actor].
2. There is no gesture (lit. band)-that cannot be used in indicating [some] idea..
3. There are besides other popular gestures (lit. hand) connected with other ideas, and they also are to be freely used along with the movements inspired by the Sentiments and the States.
4. These gestures should be used by males as well as females with proper regard to place, occasion, the play undertaken and a suitability of their meaning.

Indian Dance Forms

Bharat Muni's book Natya Shastra is the first famous source to mention dance. India has various forms of dances including classical dances and folk dances.

Dance holds a prominent place in Indian culture and is classified into two major forms:

1. CLASSICAL
2. FOLK DANCE



Classical Dance in India Classical dance traces its roots back to the ancient treatise Natya Shastra, where specific characteristics of each classical dance form are elaborated.

Folk Dance in India In contrast, folk dance evolves from the local traditions and customs of specific states, ethnic groups, or geographical regions.

Each region boasts a rich diversity of folk dances, reflecting the cultural diversity of the country.

Classical dance of India

S.No	Name of Classical Dance	Place of Classical Dance
1	Bharatanatyam	Tamil Nadu
2	Kathak	Northern India
3	Kathakali	Kerala
4	Kuchipudi	Andhra Pradesh
5	Manipuri	Manipur
6	Mohiniyattam	Kerala
7	Odissi	Odisha
8	Sattriya	Assam

CLASSIFICATION OF INDIAN DANCE

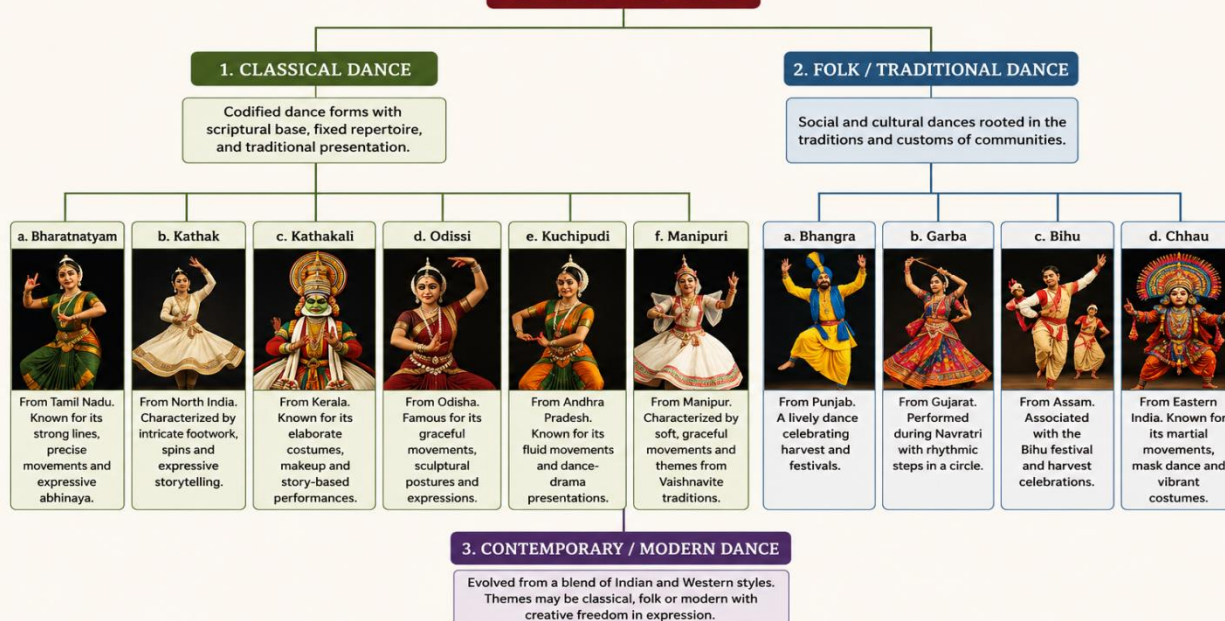
Indian dance is broadly classified into two main categories:



Note: This classification may vary slightly depending on the source, but broadly Indian dances are divided into Classical Dances and Folk/Traditional Dances.

CLASSIFICATION OF INDIAN DANCE

INDIAN DANCE



Indian Classical Dance Forms

1. Bharatanatyam (Tamil Nadu)

One of the oldest classical dances, originating from Tamil Nadu temples.

Known for fixed upper torso, bent legs, expressive hand gestures (mudras), and facial expressions. □ Usually performed solo by women as a form of devotion.

2. Kathak (Northern India)

Originated from ancient storytelling traditions (katha) in North India.

Features intricate footwork, fast spins (chakkars), and rhythmic patterns.

Influenced by both Hindu and Mughal cultures; performed by both men and women.

3. Kathakali (Kerala)

A classical dance-drama with themes from epics like the Ramayana and Mahabharata.

Recognized for elaborate costumes, painted faces, and dramatic expressions. □ Traditionally performed by men with heavy training in body control.

4. Kuchipudi (Andhra Pradesh)

A blend of dance, drama, and music, originating from village performances
Dancers often enact stories from mythology and use expressive hand gestures.
Famous for the "Tarangam" – dancing on the rim of a brass plate.

5. Manipuri (Manipur)

A graceful and lyrical dance from the Northeast, centered on themes of Lord Krishna and Radha
Known for its flowing movements, soft expressions, and traditional costumes.
Emphasizes devotional and spiritual moods rather than rhythmic footwork.

6. Odissi (Odisha)

One of the oldest surviving dance forms, inspired by temple sculpture poses
Characterized by the tribhangi posture (three bends of the body) and fluid torso movements
Depicts religious stories, mostly of Lord Jagannath.

7. Mohiniyattam (Kerala)

A graceful solo dance performed by women, named after Mohini (female form of Vishnu).
Features gentle swaying movements, feminine expressions, and elegant costumes.

Combines elements of Bharatanatyam and Kathakali.

8 . Sattriya (Assam)

Introduced by saint Srimanta Sankardeva in monasteries (Sattras) of Assam.
Strongly devotional, with themes based on Lord Krishna and Vishnu.
Performed in traditional Assamese costume with both dance and storytelling elements.

Significance of Famous Dance Forms of India	
Dance Forms	Significance
Bharatanatyam	- Bharatanatyam is one of the oldest and most widely practiced classical dance forms of India. - It preserves and showcases the cultural heritage, mythology, and spirituality of Tamil Nadu. - Through intricate footwork, expressive hand gestures, and facial expressions, Bharatanatyam tells stories and conveys emotions. - It promotes physical fitness, discipline, and body coordination. - Bharatanatyam serves as a medium of devotion and artistic expression, captivating audiences with its aesthetic beauty and cultural significance.
Kathak	- Kathak, a classical dance form originating from Northern India, combines storytelling, dance, and music. - It reflects the cultural fusion of Hindu and Muslim traditions. - Kathak is known for its swift footwork, intricate spins, and graceful movements. - It conveys emotions, depicts narratives from Indian epics, and celebrates love, devotion, and spirituality. - Kathak promotes rhythm, improvisation, and artistic expression, captivating audiences with its dynamic and expressive performances.

Kathakali	• Kathakali, a classical dance-drama from Kerala, is renowned for its vibrant makeup, elaborate costumes, and exaggerated facial expressions. • It tells stories from ancient Indian epics like the Ramayana and the Mahabharata. • Kathakali combines elements of dance, music, and acting, creating a visual spectacle for the audience. • It requires rigorous training in mudras (hand gestures), facial expressions, and intricate movements. • Kathakali preserves cultural traditions, showcases its rich mythology, and captivates viewers with its unique and dramatic storytelling.
Kuchipudi	• Kuchipudi, a classical dance form from Andhra Pradesh, combines graceful movements, intricate footwork, and fast-paced rhythms. • It originated as a temple dance and later evolved into a solo performance art. • Kuchipudi showcases mythological stories, devotional themes, and social narratives. • It combines pure dance, storytelling, mime, and music. • Kuchipudi preserves cultural traditions, celebrates regional heritage, and captivates audiences with its dynamic and expressive style.
Manipuri	• Manipuri, a classical dance form from Manipur, embodies the region's cultural heritage, spirituality, and martial arts traditions. • It features fluid movements, intricate footwork, and expressive hand gestures. • Manipuri depicts stories from the life of Lord Krishna and regional folklore. • It showcases the beauty of nature, love, devotion, and the unity of body, mind, and soul. • Manipuri dance preserves cultural traditions, fosters community identity, and captivates viewers with its grace and lyrical expressions.
Mohiniyattam	• Mohiniyattam, a classical dance form from Kerala, is known as the "dance of the enchantress." • It emphasizes feminine grace, delicate movements, and subtle expressions. • Mohiniyattam showcases themes of love, devotion, and spirituality. • It combines rhythmic footwork, expressive hand gestures, and emotive storytelling. • Mohiniyattam preserves Kerala's cultural heritage, celebrates the beauty of womanhood, and mesmerizes audiences with its lyrical and captivating performances.

Odissi	• Odissi, a classical dance form from Odisha, showcases the region's rich cultural heritage and temple traditions. • It is characterized by fluid movements, sculptural poses, and intricate hand gestures (mudras). • Odissi depicts stories from Hindu mythology, particularly those of Lord Krishna. • It emphasizes both feminine grace (lasya) and masculine vigor (tandava). • Odissi dance preserves cultural traditions, reflects devotion, and captivates viewers with its expressive and visually captivating performances.
Sattriya	• Sattriya, a classical dance form from Assam, originated in the Vaishnavite monasteries (Sattras) of the region. • It is closely associated with the religious traditions and culture of Assam. • Sattriya showcases devotion, storytelling, and mythological narratives, particularly those of Lord Krishna. • It combines graceful movements, rhythmic footwork, and intricate hand gestures. • Sattriya dance preserves the cultural heritage of Assam, promotes spiritual expression, and captivates audiences with its elegant and emotive performances.

FOLK DANCE

Folk dances are created by individuals to depict the lifestyles of people in a specific country or region. These dances are not all ethnic dances, they are not ceremonial dances or dances based on rituals. Folk dances facilitate the exploration of India's diverse cultural terrain. Folk dances, unlike classical dances, are spontaneous and performed by locals with no formal instruction and they are restricted to a small group of people or a single venue. The knowledge is passed down from generation to generation. Indian folk and tribal dances are a result of various socioeconomic conditions and long-standing traditions.

The folk dances of India reflect the diversified culture and customs of the country. There are numerous types of folk dances that are practised in various parts of the country. The folk dances of India are vibrant and full of life.

Folk dances of India are of great significance, particularly in rural areas because they mostly express the daily work and rituals of the village community.

Most of the folk dances in the country have unique costumes and they vary according to the local tradition of that particular state.

The tribal folk dances of India are inspired by the tribal folklore which is either sung by the dancers or the onlookers.

These dances are usually performed during ceremonies such as Weddings, births, coronations, entering a new house or town, welcoming a guest, religious processions, harvest time, and so on.

Some dances are performed by men and women individually while men and women dance together in other performances.

List of Folk Dances Of India (State-wise)

State	Folk Dance
Himachal Pradesh	Kinnauri, Thoda, Jhora, Jhali, Chharhi, Dhaman, Chhapeli, Mahasu, Dangi, Chamba, Thali, Jhainta, Daf, Stick dance
Punjab	Bhangra, Giddha, Daff, Dhaman, Bhand, Naqual
Haryana	Jhumar, Phag Dance, Daph, Dhamal, Loor, Gugga, Khor, Gagor
Uttar Pradesh	Nautanki, Raslila, Kajri, Jhora, Chappeli, Jaita
Rajasthan	Ghoomar, Suisini, kalbeliya, Chakri, Ganagor, JhulanLeela, Jhuma, Suisini, Ghapal, Panihari, Ginadetc
Gujarat	Garba, DandiyaRas, Bhavai, TippaniJuriun, Bhavai
Maharashtra	Lavani, Nakata, Koli, Lezim, Gafa, DahikalaDasavtar or Bohada, Tamasha, Mauni, Powara, Gouricha
Madhya Pradesh	Tertali, Maanch, Matki, Aada, KhadaNach, Phulpati, Grida Dance, Selalarki, Selabhadoni, Jawaraetc
Chhattisgarh	Gaur Maria, Panthi, RautNach, Pandwani, Vedamati, Kapalik, Chandaini, BharthariCharit, Goudi, Karma, Jhumar, Dagla, Pali, Tapali, Navrani, Diwari, Mundari, Jhumar
Jharkhand	Karma Munda, Karma, Agni, Jhumar, Janani Jhumar, MardanaJhumar, Paika, Phagua, Chhanu, Sarahul, Jat-Jatin, Karma, Danga, Bidesia, Sohrai, Hunta Dance, Mundari Dance,

	Sarhul, Barao, Jhitka, Danga, Domkach, GhoraNaach
Bihar	Jata-Jatin, Bakho-Bakhain, Panwariya, Sama-Chakwa, Bidesia, Jatra
West Bengal	PuruliaChhau, Alkap, Kathi, Gambhira, Dhali, Jatra, Baul, Marasia, Mahal, Keertan, Santhali Dance, Mundari dance, Gambhira, Gajan, ChaibariNritya
ikkim	Chu Faat, Yak ChaamSikmari, SinghiChaam or the Snow Lion, Yak Chaam, DenzongGnenha, TashiYangku, KhukuriNaach, ChutkeyNaach, Maruni Dance
Meghalaya	Laho, Baala, Ka Shad Suk Mynsiem, Nongkrem
Assam	Bihu, Bichhua, Natpuja, Maharas, Kaligopal, Bagurumba, Naga dance, Khel Gopal, TabalChongli, Canoe, JhumuraHobjanaietc
Arunachal Pradesh	Chham, Mask dance (MukhautaNritya), War dance, Buiya, Chalo, Wancho, PasiKongki, Ponung, Popir, Bardo
Nagaland	Chong, Khaiva, Lim, Nuralim, Bamboo Dance, Temangnetin, Hetaleulee. Rangma, Zeliang, Nsuirolians, Gethinglim
Manipur	Thang Ta, Lai Haraoba, PungCholom, Rakhai, Nat Rash, Maha Rash, Raukhat, DolCholam, KhambaThaibi, Nupa Dance, Raslila, KhubakIshei, LhouSha
Mizoram	Cheraw Dance, Khuallam, Chailam, Sawlakin, Chawnglaizawn, Zangtalam, Par Lam, Sarlamkai/ Solakia, Tlanglam, Khanatm, Pakhupila, Cherokan
Tripura	Hozagiri
Odisha	Ghumara, Ranappa, Savari, Ghumara, Painka, Munari, Chhau, ChadyaDandanata
Andhra Pradesh	Ghantamardala, OttamThedal, Mohiniattam, Kummi, Siddhi, Madhuri, Chhadi. VilasiniNatyam, Bhamakalpam, Veeranatyam, Dappu, TappetaGullu, Lambadi, Dhimsa, Kolattam, ButtaBommalu
Karnataka	Yakshagana, Huttari, Suggi, Kunitha, Karga, Lambi
Goa	Fugdi, Dhalo, Kunbi, Dhangar, Mandi, Jhagor, Khol, Dakni, Tarangamel, Shigmo, Ghode, Modni, Samayinrutya, Jagar, Ranmale, amayinrutya, Tonnyamell
Telangana	Perini Shivatandavam, Keisabadi
Kerala	OttamThulal, Kaikottikali, Tappatikali, Kali Auttam
Tamil Nadu	Karagam, Kumi, Kolattam, Kavadi

IMPORTANCE OF DANCE

MAIN BENEFITS OF DANCE

I. PHYSICAL

II. MENTAL/EMOTIONAL

III. SOCIAL

IV. CULTURAL

PHYSICAL BENEFITS OF DANCE

1. develops muscular and cardiovascular endurance
2. improves flexibility, coordination, balance, and body composition
3. enables joint mobility
4. helps prevent osteoporosis
5. lowers risk of cardiovascular diseases

MENTAL/EMOTIONAL BENEFITS OF DANCE

1. helps keep the brain sharp
2. decreases risk of dementia and Alzheimer's disease
3. decreases depressive symptoms
4. increases self-esteem and improves body image
5. aids in releasing emotional tension

SOCIAL BENEFITS OF DANCE

1. gives sense of togetherness within a group
2. encourages positive social interaction and interpersonal relationship in a group
3. contributes to the individual's potential for self-actualization in society

CULTURAL BENEFITS OF DANCE

1. promotes cultural values

Physical fitness;-Dancing can improve your coordination, flexibility, muscular strength, and cardiovascular health. It can also help you maintain a healthy weight and build muscle strength.

Mental health;-Dancing can help improve your mood and reduce stress by releasing endorphins. It can also help you process difficult emotions and connect with your inner self.

Cognitive function;-Dancing can improve your verbal fluency, delayed recall, and recognition memory function.

Bone strength;-Dancing can help reduce the risk of osteoporosis and strengthen your bones.

Heart health;-Dancing can help improve your endurance and balance your heart rate, which can reduce the risk of heart attack and stroke.

Self-expression;-Dancing is a creative outlet that allows you to communicate your feelings and emotions through movement.

Social skills;-Dancing can help you improve your social skills.

MUDRA

Meaning of Mudra

“Mudra” = Seal / Gesture / Symbol

In dance, it is the language of hands

Helps convey Bhava (emotion) and Rasa (aesthetic feeling)

- One of the most striking features of Indian classical dance is the use of hand gestures.
- Speaking in dance via gestures, rather than orally, in order to visually convey outer events or things, as well as inner feelings, two classifications of specific traditional 'MUDRA' (hand/finger gesture) are used in Indian Classical Dance,
- To convey inner feelings, two classifications of mudras (hand or finger gesture) are used in Indian classical dance,
- From the Natya Shastra, a text on the arts, this quotation and translation is often quoted by Indian classical dance instructors:
 1. "Yato hastastato drishtihi"... "Where the hand is, the eyes follow"
 2. "Yato drishtistato manaha"... "Where the eyes go, the mind follows"
 3. "Yato manastato bhavaha"... "Where the mind is, there is the feeling"
 4. "Yato bhavastato rasaha"... "Where there is feeling, there is mood/flavor"

In Bharatanatyam, the classical dance of India performed by Lord Nataraja, approximately 51 root mudras (hand or finger gestures) are used to clearly communicate specific ideas, events, actions, or creatures

1. `Asamyuta Hasta in which require only one hand, 28 mudras
2. 'Samyuta Hasta';- mudras which require both hands, 23 mudras

Types of Mudras

1. Asamyuta Hasta (Single Hand Mudras)

- Performed using **one hand**
- Used to show simple objects or actions

Examples:

- Pataka (flag, sky, cloud, wind)
- Tripataka (crown, tree, fire)
- Ardhapataka (leaf, river bank)

2. Samyuta Hasta (Double Hand Mudras)

- Performed using **both hands**
- Used for complex meanings

Examples:

- Anjali (salutation, prayer)
- Hamsasya (flower, delicate objects)
- Shankha (conch shell)

TEXTILE

A general term for any material made from interlaced fibers, such as yarn, thread, silk, rayon, or metal wire

History of Textile

The usage of textiles can be traced back to the Neolithic Age.

The people around 4000 BC invented hand-operated spindles and looms in Europe and the materials used were wool and flax.

The history of textiles dates back to prehistoric times, when early humans used natural fibers such as animal hair, plant fibers, and leaves to create clothing and other textiles.

These fibers were collected, cleaned, and processed by hand, and were used to create simple fabrics such as woven baskets and mats.

The processing of textiles (dyeing and printing) also has its roots in the prehistoric era.

The first solid evidence about dyeing of silk and brocades from religious and social records suggests that Indians were aware of the dyeing process in 2500 BC; however, it is also believed that the Chinese in 3500 BC practiced dyeing but no solid evidence is available to substantiate it

Safflower was in use for dyeing textiles for red and yellow colours in 2500 BC. Egyptians were able to produce a whole range of colors for textiles by 1450 BC.

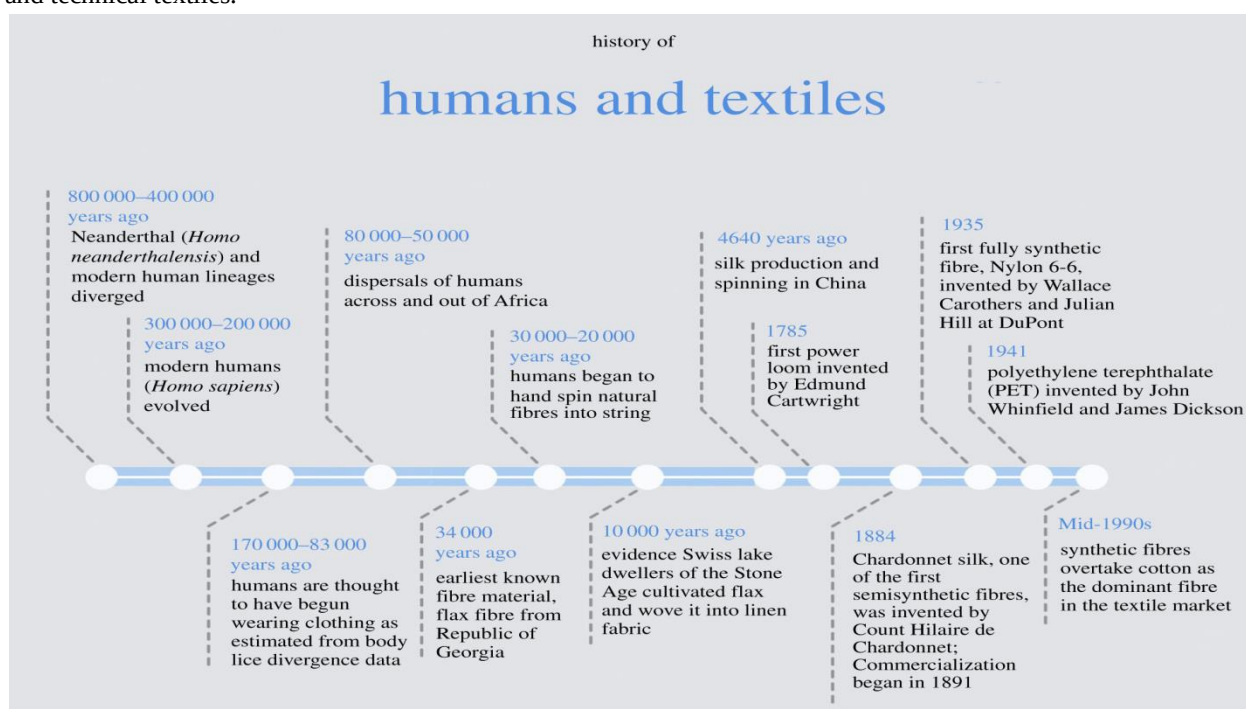
As civilizations developed, so did the techniques for producing fibers and textiles. In ancient Egypt, for example, flax fibers were used to create linen,. In ancient China, silk fibers were harvested from silkworms and used to create luxurious fabrics.

During the Middle Ages, wool became an important fiber for textiles, particularly in Europe where sheep were widely raised. Wool fibers were spun into yarn and used to create warm clothing and blankets.

Cotton fibers also became an important fiber during this time, particularly in India where cotton had been cultivated for thousands of years.

In the 18th and 19th centuries, the Industrial Revolution brought about major advancements in textile production. New machinery was developed that could spin, weave, and process fibers more quickly and efficiently. This led to the mass production of textiles, making them more affordable and widely available.

In the 20th century, synthetic fibers such as nylon, polyester, and acrylic were developed, which offered new properties and benefits over natural fibers. These fibers were used to create a wide range of products, from clothing and accessories to industrial and technical textiles.



A review of clothing microbiology: the history of clothing and the role of microbes in textiles, Volume: 17, Issue: 1, DOI: (10.1098/rsbl.2020.0700)

HISTORY OF INDIAN TEXTILES

1. Patterned cloth on stone bust, Mohenjodaro, c.2500 B.C.

India has been famous for its textiles from very early times as revealed by literary and archaeological evidence. Our knowledge of the fabrics themselves, however, is limited almost entirely to those produced after the 16th century.

2. 3000 BC

The discovery of a few fragments with traces of a purple dye at Mohenjodaro proves that cotton was spun and woven in India at least as early as 3000 B.C. We also know from a sculpture of that time that patterned cloths were in use even then

3. 5th Century BC & 4th Century BC

The Achaemenid Empire linked the Greek world to India and Darius I (521-486 B.C.) of Persia had commissioned a Greek to explore the Indus River, the Persian Gulf and the Red Sea with an eye to commerce.

4. 3rd Century BC

Chanakya, also known as Kautiliya, who was Chancellor in the Court of Chandragupta Maurya (c. 300 B.C.), mentions in his Arthashastra, that textiles were especially important among the articles of internal and external trade and has listed the important centres for cotton, silk and woolen cloth manufacturing. The materials employed in spinning were wool (urna), cotton (karpasa), hemp (tula) and flax (ksauma). Silk was imported from China. Weaving was done by women

5. 1st Century BC

There are references to the popularity of brightly coloured Indian textiles among the Persians, but it is not known when they were first traded in Europe. By the first century A.D., Indian muslins had become famous in Rome under such names as nebula, gangetika, and venti textiles ('woven winds'), the latter exactly translating the technical name of a special type of Dacca muslin. Silk was also an important export to Rome, both as yarn and as finished cloth, but the raw material was imported into India from China.

6. 1st-3rd Century AD

The 1st to 3rd century A.D. was an era of flourishing empires, four of which jointly spanned the breadth of Eurasia — the Han in China, the Kusana in India, the Parthian in Mesopotamia and the Roman in Europe. The Kusana Empire acted as the commercial intermediary for much of the trade between Rome and China. The trade in silk that passed over the Trans-Asian Silk Route must have been the main commodity, in addition to carpets, wool and furs.

7. 4th-7th Century AD

From an examination of the Ajanta paintings particularly those attributed to the 6th century A.D., it becomes possible to identify the techniques of embroidery, bandhana, (tie-and-dye); patolu (ikat, in which warp and woof are resist-dyed separately before weaving, according to the pattern required); brocading; and fine muslin weaving.

8. 13th-16th Century AD

For a study of the later mediaeval development, evidence is limited to cotton fragments discovered at Fostat and other Egyptian sites, probably of fourteenth or fifteenth century origin. Marco Polo (1254-1324) brought back to Europe valuable knowledge of India and the east. He has recorded that "Guzzerat" embroidery was regarded as the finest in the world and "all the clothes of gold and silver that are called mosolins are made in this country".

9. 16th-18th Century AD

The advent of the Mughals in India, resulting in economic prosperity and the development of a very high standard of craft workmanship. Apart from carpets, the earliest Mughal textiles known to us are of the Jahangir period (1605-1627).

TEXTILES USED IN ANCIENT INDIA

1. Cotton;-A popular choice for everyday use in India's hot and humid climate. India was the first country to grow, weave, and pattern cotton.
2. Silk;-A staple of ancient trade, with evidence of use in the Indus Valley Civilization.
3. Muslin;-An early cotton cloth that was handwoven in India and became popular in trade with Europe.
4. Organza;-A silk fabric used in saris, salwar kameezes, and churidars.
5. Kalamkari;-A textile art from South India that uses vegetable dyes and block printing or hand drawing.
6. Bandhani;-An ancient textile art form that uses a tie-dye technique to create intricate patterns and designs.
7. Georgette;-A lightweight fabric with a subtle sheen that's used to make traditional Indian sarees called "georgette pattu".
8. Linen;-A lightweight, thin fabric that's been woven from the flax plant since around 5000 B.C.

TRADITIONAL TEXTILE USED IN INDIA

Sr No	STATE	TEXTILE
1	MIZORAM	Puans: Puan-the traditional Mizo attire is primarily a wrap around skirt in white and black colour
2	HARYANA	Panja Durries: A handloom tradition that is widely used in Panipat, Haryana is Panja
3	GOA	Kunbi: The Kunbi fabric from Goa is a cotton fabric in chequered design mostly used on sarees. It was used in the olden times by tribal women
4	ODISHA	Sambalpuri Saree: Sambalpuri saree is most popular in the Odisha regions of Bargarh, Sambalpur, Sonepur, and Boudh. This is a handwoven ikat saree in which the yarn is tie-dyed primarily before being woven into a gorgeous fabric
5	SIKKIM	Lepcha: The Lepcha community in Sikkim is the most famous for its handloom tradition-the lepcha fabric. This work of art is commonly used on traditional ensembles
6	MEGHALYA	Eri Silk: Eri or Endi silk is a highly textured silk featuring short fibers compared to other varieties. It has a rich depiction of gold which makes it a valuable fabric. The silk is also used for Kantha embroidery and Pattachitra art
7	JHARKHAND	Kuchai Silk: Kharsawan-Kuchai, a Jharkhand region produces the Kuchai silk, which is an organic Indian fabric
8	HIMACHAL PRADESH	Kullu Shawls: Kullu shawls are handwoven from three wool types namely Angora, Merino, and local sheep wool. These shawls from Himachal Pradesh are commonly worn by both men and women featuring geometric patterns and bright hues
9	RAJSTHAN	Shisha: The Mughals introduced the shisha embroidery in the 17th century. Earlier, they used the mirror work embroidery on décor rather than on clothes.
10	MADHYA PRADESH	Chanderi: The traditional saree of Madhya Pradesh, Chanderi was first introduced by Shishupal, Lord Krishna's cousin. These traditional sarees are available in Chanderi cotton, pure silk, and combination of silk and cotton.

Sr No	STATE	TEXTILE
11	GUJRAT	Bandhani: The term Bandhani was derived from the Sanskrit word 'banda.' This stunning tie-dye fabric was made by Khatri, a Gujarati community, and is commonly available in shades of black, yellow, green, blue, red, and white dots that bring out interesting designs on the fabric
12	TELENGANA	Pochampally Ikat: The Ikat textile is produced from Bhodan Pochampally, Telengana and is famous for its geometric designs and intentional bleed
13	NAGALAND	Naga Shawls: Weaving is a major traditional occupation of the Naga people. The Naga shawls are highly symbolic and are made using black and red wool
14	BIHAR	Bhagalpuri Silk: Bihar is better known as the silk city. the silk weaving industry at Bhagalpur, The tussar silk is the most commonly used fabric made out of colourful threads and produced from tussar cocoons
15	WEST BENGAL	Jamdani: Jamdani is one of the main fabrics used in West Bengal. This was earlier used as a dress material for men and women but at present date, Jamdani is used in sarees with unique patterns.
16	ANDHRA PRADESH	Kalamkari: Kalamkari is derived from the Persian words, Qalam (pen) and Kari (craftsmanship). This fabric is now popularly used as a block print on most Indian attires.
17	KERALA	Kasavu: The white and gold kasavu saree is famously used by the people of Kerala especially during the celebration of Onam. Kasavu is referred to the gold zari border depicted on the mundu-saree that is made out of 100% unbleached cotton.
18	TAMILNADU	Kanjeevaram Silk: This popular silk fabric is from Tamil Nadu and is famous from their exquisite fabric and zari work. At present, the Kanjeevaram saree remains one of the popular Indian sarees with its rich depictions of gold used.

Sr No	STATE	TEXTILE
19	TRIPURA	Pachra: Pachra is the handloom fabric found in Tripura. It is a long cloth piece worn as lower attire by the women. It extends below the knee, featuring unique stripe designs and embroidery in several shades
20	UTTARAKHAND	Panchachuli Weave: The rural women of Panchachuli range in the Himalayan region commonly knit exquisite clothes made up of sheep wool and Tibetan cashmere
21	PUNJAB	Phulkari: The Phulkari work is a common embroidery work of floral depictions used on fabrics by the Punjabi women. This unique design is made on the reverse of the cloth while the design is created on the front.
22	KARNATAKA	Mysore Silk: When Tipu Sultan ruled over Mysore, the silk industry witnessed a huge growth. Till date, the state is the largest producer of silk. The rich, silk material from Mysore is famous for their quality and artistry
23	UTTARPRADESH	Chikankari: Chikankari is famously produced in Lucknow. It is a delicate embroidery design that was primarily introduced by Nur Jehan, wife of Mughal emperor, Jehangir. This cloth piece is initially block-printed and after that the chikan stitches are used on the outline
24	MAHARASTRA	Paithani: The Paithan town of Aurangabad, Maharashtra is famous for its hand-woven silk saree-Paithani.
25	CHHATTISGARH	Kosa Silk: The kosa silk is the famous fabric used in Champa district of Chattisgarh. This rich fabric is considered to be the finest in the world mostly because of its durability.
26	JAMMU & KASHMIR J&K	The shawls which are known for its inimitable tapestry design make use of a fibre commonly known among the natives as Pashm or Pashmina. The fibre normally comes from the fleece of a wild Cashmere Asian mountain goat

IMPORTANCE OF TEXTILE:

Textiles play a vital role in our everyday lives and are important for both practical and aesthetic reasons. They are versatile materials that are used in a wide range of applications and industries, and their importance is likely to continue to grow in the future.

Some of the main reasons why textiles are important include:

- 1. Clothing:** Textiles are used to create clothing and other garments that keep us warm, protected from the elements, and looking stylish.
- 2. Household items:** Textiles are used to create a wide range of household items such as curtains, bedding, and furniture upholstery.
- 3. Industrial and technical uses:** Textiles are used in a variety of industrial and technical applications such as filter fabrics, geotextiles, and protective clothing.
- 4. Health and medicine:** Textile materials are used in the production of surgical gowns, face masks, and other medical equipment.
- 5. Transportation:** Textiles are used in the production of car seats, safety belts, and airbags
- 6. Agriculture:** Textiles are used in agriculture for crop protection and to provide shade for animals

7. Leisure and sports: Textiles are used to make outdoor clothing, tents, and other outdoor gear

8.Economy: The textile industry is a significant contributor **to the global economy, providing jobs and generating income through the production and export of textile products.**

9. Creativity and Art: Textiles are used to create art, design, and fashion.

10. Sustainability: Textile production and use is becoming more sustainable through the use of recycled materials, organic fibers, and sustainable production methods.

DRESS MATERIAL

Dress material is a woven or knitted cloth made from animal, plant, or synthetic fibers that is suitable for making dresses. A specific type of textile used for clothing.

Dress materials can be made from a variety of fabrics, such as silk, velvet, chiffon, georgette, crepe, lace, cotton, or linen.

Linen; A traditional and comfortable fabric used for sarees, dupattas, and lehengas.

Silk;-A popular choice for traditional Indian dresses like sarees, Anarkalis, and lehengas.

Georgette;-A lightweight and flowy fabric used for salwar kameez and jacket style suits.

Cotton;-A comfortable, breathable, and affordable fabric used for daily wear like salwar suits, sarees, and kurtas.

Chiffon;-A lightweight, sheer, and comfortable fabric used for kurtas.

Crêpe ;-A fabric used for salwar kameez, sarees, and kurtas in both formal and casual wear.

Velvet; -A luxurious and elegant fabric used for ethnic clothing for women.

Wool;-A warm and durable fabric used for carpets, shawls, blankets, and sweaters.

HISTORY

The story of textiles in India is one of the oldest in the world and goes back to prehistoric times. Examples have been found depicting waist garments in the cave paintings of the Mesolithic era but concrete evidence of textile production and use starts appearing from the proto-historic times i.e. 3rd Millennium BCE.

ANCIENT INDIAN Clothing AND FASHION

The evidence of wild indigenous silk moth species from Harappa and Chanhudaro suggests the use of silk in the mid 3rd millennium BCE. abundance of archaeological finds and literary references.

The excavations at the site of Mohenjo-Daro (C.2500 to 1500 BCE) revealed the presence of dye vats together with woven and madder-dyed (a herb dye) cotton fragments wrapped round a silver pot.

Indus valley civilization (2600 BCE to 1900 BCE) Indus valley civilization was the pioneer of spinning and weaving cotton textiles.

VEDIC PERIOD (1500 BCE TO 500 BCE)

From the Vedic period we get references to materials and manners of dressing which point to the developments in the textile history of ancient India.

A weaver in the Rigveda is described as vasovaya. The male weaver was called vava whereas a female weaver was called vayitri.

The Rigveda refers mainly to two garments, vasa or the lower garment and the adhivasa or the upper garment. In the later samhitas, the nivi or undergarment had also come into use.

THE POST VEDIC PERIOD witnessed two divergent lines in respect of the nature or source of materials used for textile purpose. Those pursuing Vedic line of animal sacrifice sought the major source of clothing in skins, while others, the Buddhists and Jains in particular, in cotton.

THE SANSKRIT EPIC MAHABHARATA (C. 400 BCE-C. 400 CE) mentions silk fabrics among the presents brought to Yudhishthira by feudatory princes from the Himalayan regions. References of weaving are very common from Rigveda onwards. There were female weavers from earliest times.

ACCORDING TO THE VALMIKI'S RAMAYANA(5TH/ 6TH- 3RD CENTURY CE), the trousseau of Sita consisted of woolen clothing, furs, precious stones, fine silk vestments of diverse colours, princely ornamental and grand carriage of different kinds. The veil, bodice, and body clothes are repeatedly mentioned in the epics Ramayana and Mahabharata.

MAURYAN & SUNGA (321 BCE TO 73 BCE) ;-men and women in mauryan and sunga dynasty wore unstitched garments, with the main garment being the antariya made of white cotton, linen or muslin

KUSHAN (1ST TO 3RD CENTURY CE) ;-adaptation of stitched garments. central asian styles on indian clothing

GUPTA (320 CE TO 550 CE) ;-significant improvement in textile printing, embroidery, and dyeing techniques, which resulted in high-quality fabrics and intricate designs

MUGHAL CLOTHING refers to clothing worn by the Mughals in the 16th, 17th and 18th centuries throughout the extent of their empire. Much of them were already being used in the past centuries before their arrival in Indian subcontinent. It was characterized by luxurious styles and was made with muslin, silk, velvet and brocade. Elaborate patterns including dots, checks,

and waves were used with colors from various dyes including cochineal, sulfate of iron, sulfate of copper, and sulfate of antimony were used.

During the British period in India, the primary dress materials were cotton, silk, mill-made cloth, and other imported satins and artificial silks:

Cotton; India exported cotton to Britain, and the British imported mass-produced garments made from it.

Silk;-Soft-washing silks, silk flannel, pongee silks, and black silk dresses were popular.

Mill-made cloth;-This was cheaper to produce than the cloth used in India and was more versatile.

Chintz;This was a light, color-fast cotton fabric made in India for the English market. The original chintz designs were hand-painted, but block-printed designs were later incorporate

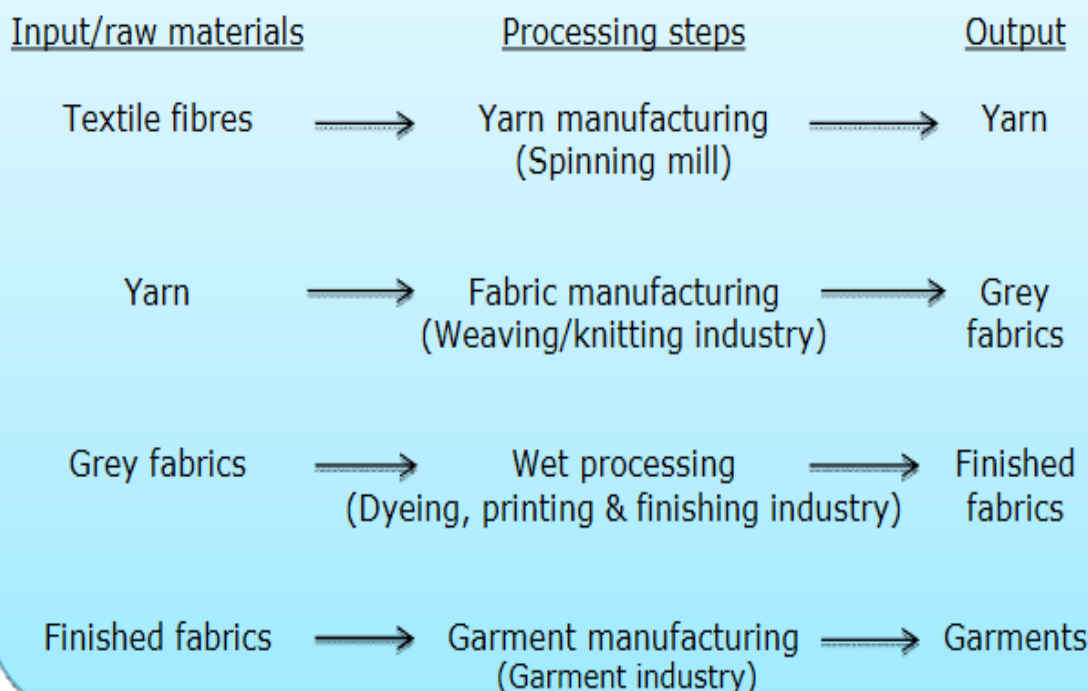
Ancient Indians used a variety of materials to make their clothing, including:

1. Cotton: India was one of the first places to cultivate cotton, and it was a primary material for clothing.
2. Silk: A significant material for the wealthy, silk was used in the uttariya, a scarf that draped the top half of the body.
3. Cotton –silk
4. Linen
5. Cotton –linen
6. Silk linen
7. Hemp: An affordable material for weavers and laborers.
8. Wool: Used in the form of blankets during the winter in the warmer regions of India.
9. Flax: A plant grown for its fibers.

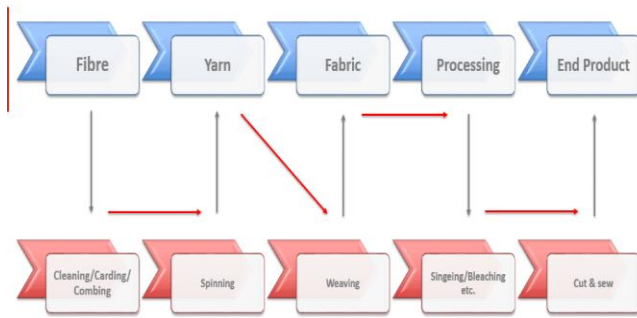
In ancient India, people used a variety of materials to make their clothing, including:

1. **Cotton;** -First found in India around 600 BC, cotton was widely used to make clothing. Women wore cotton sarees, while men wore lungis.
2. **Silk;**-A popular choice for traditional Indian clothing, silk is derived from the fibers of silkworms. It's used to make sarees, lehengas, and anarkalis.
3. **Linen;**-A soft and light fabric made from the flax plant, linen was used to make clothing for ancient Indian kings.
4. **Muslin;**-Made from locally grown cotton called Kapas, muslin was a legendary cloth of East India. Muslin from the eastern parts of ancient India was highly valued in the international market.
5. Animal skin and fur;-In **ancient times, people also used animal skin and fur to make their clothing.**
6. **Tree leaves;**-People also used tree leaves to make their clothing.
7. **People in ancient India also wore jewelry made of gold, silver, copper, and stones like lapis lazuli, turquoise, amazonite, and quartz.**

Flow chart of textile processing:



COTTON TEXTILE MANUFACTURING PROCESS



STEPS INVOLVED IN MAKING COTTON FABRIC FROM COTTON BALLS

The process of manufacturing cotton fabric from cotton balls started with cotton cultivation. And this process ends with weaving cotton fabric. Now here in this section of this article, we'll go through all the processes involved with manufacturing cotton fabric from cotton balls.

Step-01: Cultivation of cotton

The cotton is planted in the early spring. In around 60 days, the plants begin to blossom, producing whitish-yellow flowers. These flowers mature into circular walnut-like structures throughout time. Cotton balls are what they're called. Cotton fibers sprout on these grains. Green cotton balls eventually become brown. Cotton bolls break open at maturation, revealing the white cotton fiber.

Step-02: Ginning

Cotton that has been harvested from the plants will have seeds embedded inside it. Ginning is the term given to the process of extracting cotton seeds from the pods. Hand labor was historically used in the process of ginning. Ginning is done with the assistance of machinery these days.

Step-03: Spinning

Spinning is the very next process to make cotton fabrics from cotton balls. Producing cotton yarn from the cotton fiber is called spinning. The spinning process is involved 3 easy steps.

Step-3.1: The raw cotton is unfastened and washed to eliminate straw as well as dried leaves.

Step-3.2: After that, the cotton is washed and put into a machine. Then cotton fibers are combed, straight, as well as turned into a sliver, a rope-like form.

Step-3.3: The lint-like fragment of cotton fiber is transformed into yarn via the spinning process performed by machines.

Step-04: Dyeing

At this step, the basic material utilized in cotton garments is finalized. This cotton yarn may subsequently be exposed to a number of chemical treatments as well as this yarn may be dyed.

Step-05: Weaving

Weaving refers to the act of making cloth by interlacing two different sets of yarn in a particular pattern. Bobbins are the common name for the large reels of yarn. The weaving of the fabric requires the employment of these bobbins

1. Cotton Cultivation & Harvesting

- Cotton plants are grown in fields
- Mature cotton bolls are handpicked or machine harvested
- Raw cotton contains seeds, dust, and fibers

2. Ginning

- Fibers are separated from seeds using a **ginning machine**
- Produces clean cotton lint
- Seeds are sent for oil extraction or other uses

3. Spinning

- Cotton fibers are cleaned, carded, and drawn into slivers
- Slivers are twisted into **yarn**
- Yarn is wound onto bobbins or cones

4. Weaving / Knitting

- Yarn is converted into fabric

Weaving:

- Two sets of yarn (warp + weft) interlaced on a loom

Knitting:

- Yarn is looped using machines to form fabric

5. Dyeing & Printing

- Fabric is colored using dyes
- Designs are printed using different techniques
- Can be done at yarn stage or fabric stage

6. Finishing

- Improves texture, strength, and appearance
- Processes include:
 - Bleaching
 - Mercerizing
 - Softening
 - Shrinking control

7. Cutting & Garment Manufacturing

- Finished fabric is cut into patterns
- Sewn into garments like shirts, pants, etc.

WEAVING OF COTTON

Cotton weaving is the process of interlacing two sets of yarn, the warp and weft, to create fabric. The process takes place in the latter stages of cotton manufacturing. Weaving is a process in which two sets of yarns are weaved together to make a fabric. The lengthway threads, known as the warp and cross way threads known as the weft, are interlaced on a loom to form the desired fabric.

Classification

1. Hand weaving
2. Machine weaving

Weaving process

Flow Chart of Weaving

Yarn from spinning section

Doubling and Twisting

Winding

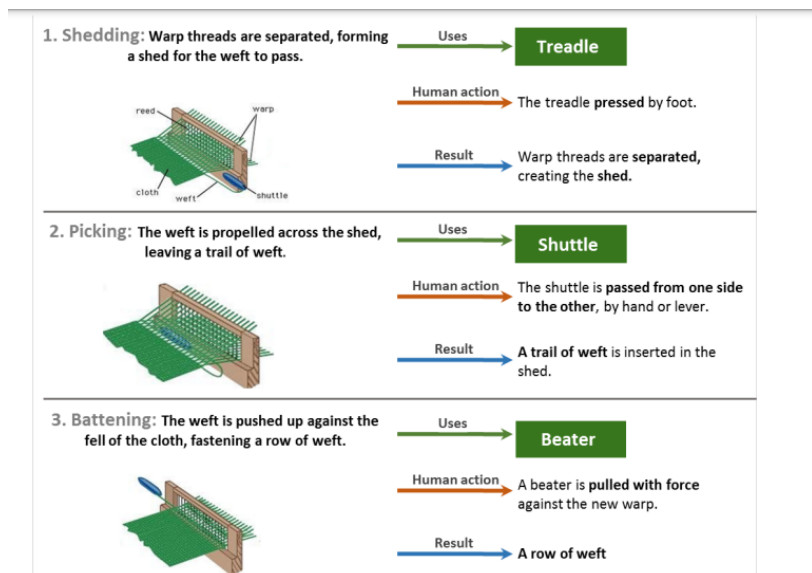
Creeling

Warping

Sizing

Winding on weavers beam

Weaving



Primary Motion of Weaving: In order to interlace warp and weft threads to produce fabric on any type of weaving machine, three operations are necessary:

- A. Shedding: Separating the warp threads, which run down the fabric into two layers to form a tunnel known as the shed.
- B. Picking: Passing the weft threads, which traverses across the fabric, through the shed.
- C. Beating-up: Pushing the newly inserted length of weft, known as the pick, into the already woven fabric at a point known as the fell.

DYEING OF COTTON

Flow Chart of Dyeing



Dyeing is an ancient art which predates written records and it was practiced since Bronze Age¹. In textile processing dyeing is an integral part where textile coloration is done to make the fabric lively. Dyeing is the process of adding color to textile products like fibers, yarns, and fabrics.

There are mainly two classes of dye:

1. Natural
2. Man-made(chemical)

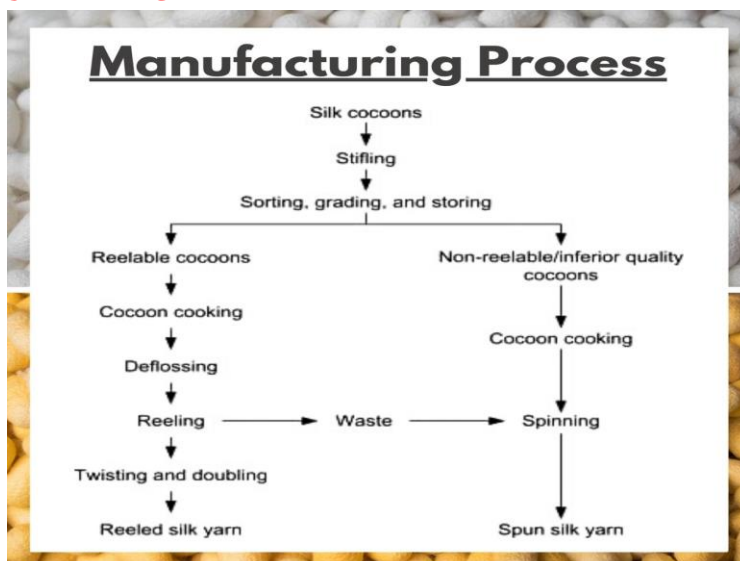
Natural Dyeing Methods - Natural Many companies are becoming more environmentally friendly and are using more natural dyes. These natural dyes are what used to be used years ago before the advent of the chemical processes. These work better with natural or regenerated fabrics and they require a mordant to fix them. It is difficult to reproduce the same colour each time. This technique is particularly good for:

- Natural fabrics
- Small quantities Dyeing & Printing

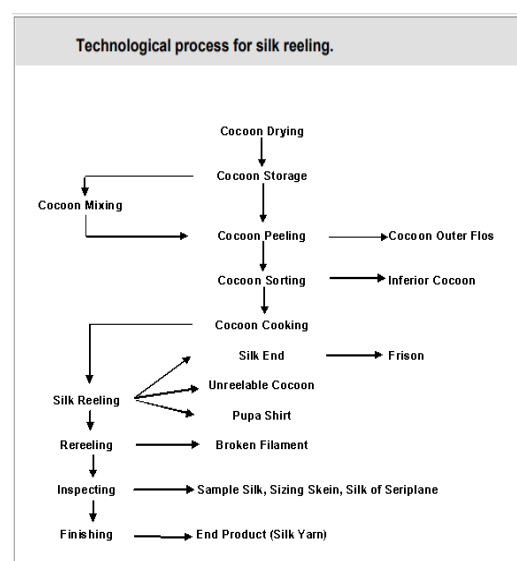
Man-made Dyeing Methods - Chemical Chemical dyeing uses pigment dyes with salts to fix them. Fabric is immersed in the dye bath until the right depth of colour is achieved. This is the sort of dyeing that can be done at home. The best sort of fabrics to take up the dyes are things like cottons and silks – natural fabrics. There are cold water dyes available, and also dyes that can be put into the washing machine with the fabric to be done. The machine is then set to a wash cycle and the fabric dyed. This technique is particularly good for:

- Natural fabrics
- Small quantities Dyeing & Printing

SILK FABRIC



SILK WEAVING PROCESS



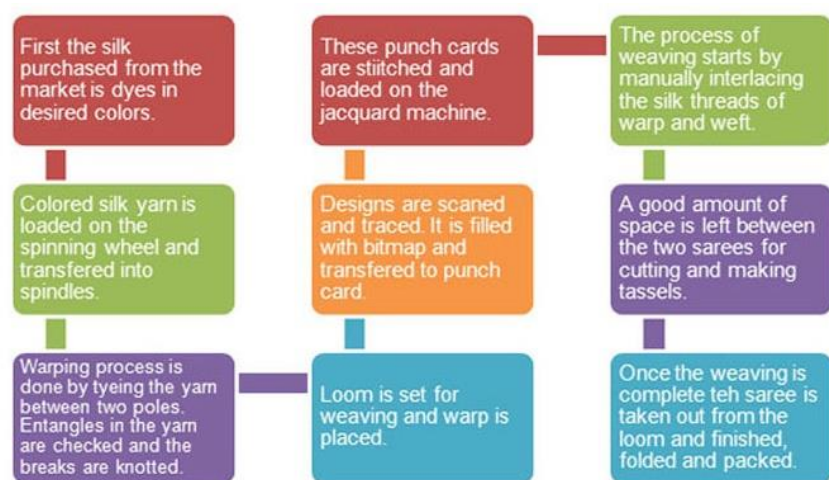
Silk weaving is the process of interlacing two sets of threads to create a fabric. The threads are woven at right angles to each other, with one set running up and down the fabric (the warp) and the other set running across it (the weft). The type of weave used determines the finish of the silk. Some common types of silk weaves include plain weave, satin weave, and open weave.

Here are some steps involved in the silk weaving process:

Loading the yarn: The yarn is loaded into the loom as the warp.

Weaving: The weaver uses their hands and legs to interlace the warp and weft threads. The weaver operates a pedal to pass a shuttle through the openings formed by the interlocked threads. Once the shuttle passes, the suspended sley is pulled to form the weave

Weaving is done on the fly shuttle pit looms. The weaver interlaces the silk threads of weft and warp. The shuttle passes through the opens formed when the pedal is operated to interlock the threads of warp and the weft. Once the shuttle is passed, the suspended sley is pulled to form the weave. The portion of woven cloth is wound to the wooden beam which is located in front of the weaver. After weaving of 6 yards of weft, the portion of unwoven warp is intentionally left before and after the sari weaving which is later knotted for fringe. Thus the weaving is completed- the sari is smoothened using brass metal blade and sari is folded in traditional manner for the market. It takes nearly 4 to 5 days to complete one sari. The length of 10 saris warp is loaded into the loom at a time. The weaver may need 1 or 2 persons to help him while working.



Silk Dyeing:

Silk being a natural polyamide or polypeptide fiber, its dyeing properties are very similar to those of other natural polypeptide fiber, wool as well as synthetic polyamide fiber, nylon. They can be dyed by similar methods. Very fine fibrillar structure and high orientation of fiber molecules are the two characteristic properties of silk which determine its dyeing behavior.

silk is dyed using various dyes such as Acid dyes, metal-complex, reactive dyes etc. Acid dyes are more suitable for silk and wool.

Pretreatment Before Dyeing

Degumming Of Silk;-The process of eliminating sericin (gum) is known as degumming.

Scouring;-a pretreatment step in textile production that removes impurities from fabrics to prepare them for dyeing

Bleaching; Bleaching of silk is carried out to remove natural colouring matter of silk

Fluorescent Whitening;-uses fluorescent whitening agents (FWAs), also known as optical brightening agents (OBAs), to make materials appear whiter

Step 1: BoM(Bill of Materials-Acid Dyes ,Hot water, Measuring cylinder,Scale, Synthrapol(or similar textile detergent),Large jars , Soaking container, Masking tape & markers (for labeling), Coffee stirrers/stirring sticks,Rubber gloves, protective wear, goggles etc

Step 2: SynthrapolSoak;-All fiber has to be soaked before it can be dyed. Soaking allows the fibers to open up, making it easier to absorb dye. Silk can be hard to wet so the longer you let it soak, the more successful your dyeing will be.

Step 3: Dye Preparation &Soak ;-How much dye powder you will need depends on how much fiber you are dyeing.Add the mixture to your dye jar and then add fiber. Add more water to your jar to allow the silk to move around more freely. Let the silk soak in the dye bath for at 12-24 hours.

Step 4: Add Heat ; - After the 12-24 hour dye soak, slowly turn up the heat. Don't let the water go over 180 or get to a boil. Too much heat will cause the silk to lose luster.

Step 5: Cool Down &Rinse;-Once the dyes have all allow to cool down overnight.To rinse the silk, remove it from the dye bath (which will likely be very clear) and rinse under cool water until the water runs clear.

Step 6: Dry;-Allow the silk to air dry before use.

The silk dyeing process involves several steps, including:

1. Checking the fabric: Before dyeing, it's important to confirm that the fabric is 100% silk. Other fibers may not absorb the dye evenly or react differently. You can check the fabric content label or perform a burn test.
2. Preparing the dye: The dye is pasted with water and heated to 40°C.
3. Adding the dye: The dye is added to water with the silk in it.
4. Dyeing: The dyeing process takes place in the presence of acid at a temperature of up to 60°C for 2–4 hours.
5. Squeezing and drying: The silk is squeezed and dried

ANCIENT DRESS MATERIALS AND TEXTILES RELEVANCE IN TODAY CPONTEXT

The legacy of ancient dress materials and textile traditions is highly relevant in the contemporary world in several ways:

Cultural Identity and Heritage: Traditional garments like the Indian sari, Japanese kimono, and African Kente cloth are vital expressions of cultural identity and heritage that continue to be worn for everyday life, ceremonies, and festivals today.

Sustainable Fashion: With growing concerns about the environmental impact of "fast fashion" and synthetic fabrics (like polyester and nylon), there is a significant resurgence of interest in sustainable and eco-friendly natural fibers (organic cotton, linen, silk, hemp) and natural dyeing practices.

Inspiration for Modern Design: Contemporary designers frequently draw inspiration from traditional fabrics, ancient motifs, patterns, and historical silhouettes, incorporating them into modern fashion lines.

Economic Empowerment: The preservation and promotion of traditional handloom and handicraft techniques create employment opportunities, particularly in rural areas, and support local artisan communities, boosting the textiles and apparel sectors in many countries.

Technological Innovation: The fundamental techniques developed in ancient times (weaving, knitting, spinning) are the basis of modern textile technology. Today's innovations focus on enhancing the functionality of these materials, making them water-proof or UV-resistant, and using computer-assisted design (CAD) software while still honoring traditional methods.

WEAVING AND DYING RELEVANCE IN TODAY'S CONTEXT

Relevance of Weaving

- **Cultural Preservation and Identity:** Weaving techniques are deeply rooted in the history and cultural identity of communities worldwide. By continuing these practices, communities preserve traditional knowledge and a sense of identity, connecting current generations with their heritage.
- **Economic Empowerment:** Handloom weaving provides a vital source of income for millions of artisans, particularly in rural and marginalized communities and for women, contributing significantly to local economies and sustainable development.
- **Sustainable Fashion:** In contrast to the environmental toll of fast fashion, traditional weaving methods often use natural fibers and low-impact, energy-efficient production techniques, aligning with the growing consumer demand for eco-friendly and ethical clothing choices.
- **Aesthetics and Uniqueness:** Handwoven fabrics are valued for their authenticity, unique textures, and imperfections that cannot be replicated by machines. This appeals to designers and consumers looking for exclusivity and a personal connection to the product's story.
- **Innovation and Technology:** Modern advancements, such as digital weaving software and e-commerce platforms, are being integrated with traditional methods. This hybrid **approach helps artisans access global markets and allows for design innovation while preserving the core craft.**

Relevance of Dyeing

- **Environmental Concerns:** The textile industry's use of synthetic dyes is a major source of water pollution and chemical waste. This has spurred a significant revival of natural dyeing as a greener, biodegradable alternative.
- **Health and Safety:** Natural dyes derived from plants, fruits, and minerals are often non-toxic, making them safer for both textile workers and consumers, reducing the risk of skin irritation and environmental harm.
- **Functional Properties:** Many natural dyes offer inherent functional benefits to textiles, such as antimicrobial, antioxidant, and UV-protective properties, which are desirable features in modern performance wear and home goods.
- **Ethical Sourcing and Transparency:** The use of natural dyes and transparent, traceable supply chains addresses consumer demand for ethical practices. Brands that highlight the natural and responsible processes behind their dyed textiles build trust and brand loyalty.
- **Artistic Expression:** Artists and designers are using both traditional and innovative dyeing techniques to create unique, artistic textiles that push creative boundaries and offer new ways of experiencing fabric purely as art.